The Role of Private-Voucher Schools in the Education-to-Work Transition*

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November 2, 2026

Abstract

We study how private-voucher high schools shape the transition from school to work in Chile. Using administrative data that follow a cohort of public-primary eighth graders into higher education and the labor market, we estimate overall and sector-specific returns with schooling, and ask whether vouchers are associated with higher returns to various schooling levels, and whether vouchers induce different schooling paths, relative to public high schools. We find that what higher education pays depends on the type of program as much as on completing it. A professional degree raises wages by 0.36 log points relative to a high-school diploma; a technical degree raises wages by 0.24 log points. Conversely, an incomplete professional spell lowers wages by 0.23 log points, while an incomplete technical spell raises wages by 0.06 log points. Because completion is far from assured, the expected return to enrolling in a professional program is statistically indistinguishable from zero, against a technical program's 0.14 log points, and it carries substantially more risk. Attending a private-voucher high school raises higher-education enrollment by 2 percentage points, and the increase is entirely professional. However, most of that increase materializes in dropout. Yet wages rise by 0.06 log points. We conclude that essentially all of the positive effects of vouchers on labor market outcomes come from higher returns within schooling states.

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1 Introduction

For decades the promise of higher education has been treated as nearly automatic. Enroll, and a labor-market payoff follows. However, recent studies have shown that this promise may no longer be a panacea (Stange, 2012; Rodríguez et al., 2016; Heckman et al., 2018; Mountjoy, 2022; Zühlke et al., 2022). Enrolling does not guarantee a return, because a large share of those who enroll never finish, and an incomplete spell can carry a low or even negative labor-market return. As a consequence, a policy that succeeds in pushing more students into higher education need not improve their outcomes. Part (or all) of the additional enrollment may end in dropout, and the programs students are channeled into may be the ones in which an incomplete spell costs most. Evaluating such a policy only by the enrollment it generates therefore risks misreading both whether it works and why. A more complete evaluation should include degree completion and the associated labor market returns.

In this paper, we study how private-voucher high schools shape the transition from school to work. Despite the voucher literature having provided plentiful evidence on the effects of voucher schools on test scores (Rouse, 1998; Angrist et al., 2002; Mayer et al., 2002; Peterson et al., 2003; Wolf et al., 2013; Lara et al., 2011; Muralidharan and Sundararaman, 2015; Epple et al., 2017; Mills and Wolf, 2017; Abdulkadiroglu et al., 2018), far less is known about what vouchers do further along the educational sequence, where the decisions that determine labor-market outcomes are actually made.[1] We follow a cohort of Chilean students from the end of primary school into the labor market and ask whether the high school they attend changes whether they continue past secondary education, which kind of higher-education program they enter, whether they complete it, and what they earn.

Chile is an interesting scenario to study vouchers. With its more than forty years of experience with a nationwide voucher program, it constitutes one of the world's most important large-scale experiments of educational choice (Epple et al., 2017).[2] Since its inception in 1981, the basic aspects of the system have remained fairly unchanged, with students exercising unrestricted choice among public and subsidized private schools.[3] Each student is entitled to a voucher that

[1] There is a small, but growing, literature that has analyzed the effects of vouchers on longer-term educational outcomes, mainly high school graduation and college enrollment (Angrist et al., 2006; Bettinger et al., 2010; Wolf et al., 2013; Chingos and Peterson, 2015). Fewer studies follow voucher recipients into the labor market. For instance, Bettinger et al. (2019) track applicants to Colombia's PACES lottery for more than twenty years and find that the program's gains—7 p.p. in tertiary enrollment and 17 percent in formal-sector earnings at about age 33—come entirely from students who had applied to vocational secondary schools.

[2] Gauri (1998) describes Chile's voucher system as "...perhaps the world's most ambitious attempt to design and implement a program of educational choice...".

[3] In 2008, the Chilean government introduced an important reform to its voucher system, that, among other things, created a differentiated voucher for low-income students. However, since the secondary-school choice we study precedes this reform, we do not go into the details of the implied changes in the system's structure. See Correa et al. (2014),

is directly paid to the school of her choice, and that covers all or part of the school's tuition. Schools, on the other hand, use the vouchers they receive to fund their operations.

Our empirical analysis describes a specific and policy-relevant population. We follow the 94,430 students who completed eighth grade in 2000 in a public primary school offering no secondary grades and who were therefore required to choose a new high school—about two-fifths of the national eighth-grade cohort and two-thirds of its public-primary students—throughout the schooling choices they make until they leave the educational system and into their early years in the labor market in 2013. These students score about 0.14 standard deviations below the national eighth-grade mean and are about 1.3 times as likely as the average Chilean student to have a mother who did not complete secondary school (58% versus 45%). They are, in other words, the students a voucher system is meant to serve, and the forced choice they face at the end of primary school is what makes the secondary-school decision well defined for all of them.

We combine two complementary empirical approaches. We begin with a set of reduced-form value-added regressions, building on the value-added literature (Rockoff, 2004; Rivkin et al., 2005; Kane and Staiger, 2008; Chetty et al., 2014a,b; Bau, 2022), that project each labor-market outcome on the student's final schooling level, or on the type of high school attended, controlling for the four standardized tests the student sat in eighth grade. The identifying assumption is the usual one in the value-added literature, namely that conditioning on prior achievement absorbs the ability that drives both schooling choices and later outcomes. As a consequence, these regressions recover average treatment effects cleanly and with little structure. What they cannot deliver is who gains and who loses, or what would happen under counterfactual assignments, both of which are central questions for policy. We therefore develop and estimate a sequential model of schooling decisions and outcomes that incorporates the decision stages students face as they move through the system, beginning with the choice of a private-voucher or a public high school. The model is a generalization of the Roy model (Roy, 1951; Heckman and Honoré, 1990), in which individuals compare the net benefits of the alternatives at hand. We allow observed and unobserved characteristics to influence both decisions and outcomes, and our identification strategy, which recovers the distribution of unobserved heterogeneity from a system of early test scores, supports interpreting that heterogeneity as a combination of inherent scholastic abilities (Carneiro et al., 2003; Hansen et al., 2004; Heckman et al., 2006b, 2018). The model also delivers the potential outcomes each student would realize on paths she does not take. Reassuringly, the value-added and structural estimates of the wage returns to each schooling level, and of the vouchers' effects on wages, are closely aligned.

Three findings organize the paper. First, what higher education pays depends on the type

Navarro-Palau (2017), Feigenberg et al. (2019), Gazmuri (2024), Neilson (2025), and Sánchez (2026) for studies that analyze this reform.

of program a student enters as much as on whether she completes it. Relative to a high-school diploma, a professional degree raises wages by 0.36 log points and a technical degree by 0.24, but dropping out of a professional program lowers wages by 0.23 log points and formal employment by 9.8 percentage points (p.p.), whereas dropping out of a technical program raises formal employment by 2.2 p.p. and wages by 0.06 log points. Interestingly, both degrees leave formal employment lower than a high-school diploma, by 4.9 p.p. for a professional degree and 1.5 p.p. for a technical one—which is possibly due to the accumulation of experience effect of high school degree holders having entered the labor market earlier than higher education degree holders (Heckman et al., 2006a). The expected return to enrolling in a professional program is 0.02 log points, statistically indistinguishable from zero, against 0.14 log points for a technical program. The professional option also lowers the probability of formal employment by 7.7 p.p., and its outcome distribution is markedly more dispersed than that of enrolling in a technical program. As a result, the technical track dominates the professional one in expected return, employment and risk, and its advantage in wages and in employment holds in every ability tercile.

Second, private-voucher schools send students to the professional track. Attending a private-voucher high school raises higher-education enrollment by 1.97 p.p. out of a base of 46.7%, and the entire increase is professional—professional enrollment rises by 2.01 p.p. while technical enrollment does not move—and it is drawn out of non-enrollment. However, most of it does not materialize in a degree. Professional dropout rises by 1.37 p.p. and technical dropout by 0.72 p.p., while professional graduation rises by 0.64 p.p. and technical graduation falls by 0.77 p.p., so the additional enrollment leaves overall degree attainment no higher.

Third, we estimate private-voucher schools' effects on labor market outcomes and decompose those effects into the schooling flows they induce and the returns those destinations pay. Vouchers raise wages by 0.06 log points. Importantly, essentially all of that effect runs through higher wages within each schooling state, and virtually none through transitions across these states. The within-state component accounts for 0.06 log points, while the reallocation across schooling states contributes -0.001, statistically indistinguishable from zero.

This paper contributes to the literature in three ways. First, it adds to the studies that estimate the effects of school-choice programs on schooling outcomes beyond test scores. Past studies have considered high-school graduation (Angrist et al., 2006; Bettinger et al., 2010; Wolf et al., 2013), college enrollment (Chingos and Peterson, 2015; Chingos, 2018), degree attainment (Deming et al., 2014; Wolf et al., 2019; Yoo et al., 2023; Cohodes and Pineda, 2024), and the two-year/four-year college margin (Angrist et al., 2016; Dobbie and Fryer, 2020; Cohodes and Pineda, 2024). Absent in these studies is the incomplete educational spell as a terminal state. On that line, Wolf et al. (2019) show that Milwaukee voucher users enter four-year colleges at

a higher rate than students with no voucher, but degrees do not move in the older cohort.[4] We extend these studies by observing every schooling step for the same voucher-treated students and treating the incomplete spell as a terminal state. Angrist et al. (2016) anticipate the form in Boston, where charters move applicants from two-year to four-year colleges while leaving overall enrollment statistically unchanged. In our case, enrollment itself rises, from non-enrollment into the track we find dominated, and arrives mostly as professional dropout.

Second, we add to the small literature on the effects of voucher and charter schools on earnings (Bettinger et al., 2019; Dobbie and Fryer, 2020; Yoo et al., 2023), and on how schooling attainment and earnings effects relate (Chetty et al., 2014b; Deming et al., 2016). The two effects often diverge. Chile's original voucher reform raises every attainment margin yet leaves mean earnings virtually unchanged (Bravo et al., 2010). Similarly, charters raise four-year enrollment, but not earnings (Dobbie and Fryer, 2020). We contribute to these studies by identifying the mechanisms that translate schooling attainment into earnings, by splitting a sector's earnings effect between the postsecondary states it moves students across, each at its own return, and the change in pay within them. Our evidence points at the induced schooling carrying no measurable value for wages, while the wage effect itself is positive. Among the studies we are aware of, none reports this configuration.

Third, we add to the literature on the returns to starting higher education when completion is uncertain. Studies in this literature estimate the effects of starting college taking into account completion (Altonji, 1993), the option to stop (Stange, 2012), each node's continuation value (Heckman et al., 2016, 2018), and vocational against university entry net of failure risk (Zühlke et al., 2022), with state-specific returns net of ability (Carneiro et al., 2003; Heckman et al., 2006b; Urzua, 2008; Prada and Urzúa, 2017) and "some college" as a terminal state (Heckman et al., 2006b; Urzua, 2008). Among US studies, most of them find that the incomplete state pays above a high-school diploma, so the continuation value only adds to a positive base. An exception is Altonji (1993), who shows that leaving with under two years of college is likely and pays -0.61 percent for men, whose ex-ante return to starting is 2.8 percent.[5] We extend this literature by computing the returns to entering each track, accounting by student's own completion, against a common high-school baseline and net of unobserved ability, and find a ranking. The professional track is dominated in wages, employment and risk, its negative direct

[4] In a similar vein, Bettinger et al. (2019) follow PACES applicants to about age 33, and find that the 7 p.p. tertiary-enrollment gain among vocational-school applicants comes with 2.4 p.p. more tertiary graduates. In the US, Massachusetts charter schools raise four-year enrollment by 8.2 to 9.7 p.p. and bachelor's completion by 4.1 to 11.4 p.p. (Cohodes and Pineda, 2024).

[5] In Heckman et al. (2016), a quarter of the return to enrolling in college comes from the continuation value, the option it opens to graduate. Rodríguez et al. (2016) estimate the returns to Chile's two-, four- and five-year degrees, but exclude dropout by construction. Prada and Urzúa (2017) estimate the returns to four-year attendance, finding that 19 percent of high-school graduates would earn more by not attending.

effect leaving the continuation value as its entire case. Our findings also relocate the heterogeneity. We show that the degree return is flat in ability while the dropout penalty steepens.

This paper is organized as follows. Section 2 describes Chile's voucher system and the facts that motivate our analysis. Section 3 presents the data and the construction of our sample. Section 4 reports the value-added evidence on what each schooling level pays and on the effect of attending a voucher high school. Section 5 presents our model of schooling decisions and outcomes, and discusses its identification and estimation. Section 6 presents the empirical implementation. Section 7 presents our results. Finally, Section 8 concludes.

2 Institutional Background

Chile has run a nationwide school voucher system since 1981. Families choose schools freely, unconstrained by residence and subject only to the distance they are willing to travel and to any top-up fees a school charges, and funding flows from the government to schools through a per-student voucher. Schools fall into three groups by ownership, management, and subsidy status. *Public* (municipal) schools are owned and managed by local municipalities, are funded by the voucher, and are tuition-free. *Private-voucher* schools are privately owned and managed, receive the subsidy, and may charge top-up fees. *Private-non-voucher* schools are privately owned and managed, receive no subsidy, and are financed entirely through fees charged to parents, that are an order of magnitude higher than the top-up fees families may pay at private-voucher schools (Sánchez, 2026). In 2004, the year our cohort completed secondary school, among the private-voucher secondary schools that charged top-up fees, the average fee was about CLP 14,000 or US$23 per month. In the early 2000s, 46% of students attended a public school, 45% a private-voucher school, and 9% a private-non-voucher school.[6]

We follow standard practice and set aside the private-non-voucher sector, which does not take up vouchers, applies selective admissions, and lies outside the public-versus-subsidized school-choice framework (Contreras et al., 2010; Lara et al., 2011; Correa et al., 2014). Our treatment contrast is therefore the binary choice between a private-voucher and a public high school. We study that choice among students who completed eighth grade in a public primary school offering no secondary grades.[7] These students must choose a new school at the end of primary education, which makes the secondary-school decision observable and well defined for the entire sample.[8]

Higher education in Chile is frequently begun and often not finished. Among the students in

[6]For more detailed descriptions of Chile's universal voucher system, see Gauri (1998), McEwan and Carnoy (2000), and Gazmuri (2024).

[7]In Chile, primary education comprises eight grades, while secondary education goes from ninth through twelfth grades.

[8]This is in contrast to the inertial decision faced by students whose primary school continues into secondary.

the broad population who enter higher education, about half leave with a degree. Dropout is high on both professional and technical tracks. On the professional track—the academic, university-oriented programs, which are also the longer ones—47% of enrollees leave without a degree, and on the technical track of shorter vocational programs 54% do. Enrollment alone is therefore a poor summary of what higher education delivers.[9]

Voucher students attain more than their public-school peers. They enroll in higher education at a rate of 66% against 51%, and the whole gap sits on the professional track, 45% against 30%, while technical enrollment is the same in both sectors, 21%. They also leave with more degrees, 33% against 24%, and conditional on enrolling they graduate somewhat more often, 50% against 48%. This picture is consistent with a generally encouraging body of Chilean evidence, which has tended to find positive effects of the voucher sector on standardized test scores (Contreras et al., 2010; Lara et al., 2011).

3 Data

Our sample consists in the universe of 8th grade students enrolled in public primary schools that do not offer secondary grades, in the year 2000. The majority of these students are 14 years old by then. We follow these students through high school. Once graduated from high school, we observe whether they enter higher education over the seven years that follow, so that a student counts as an entrant only if she first enrolls by 2011. For students who do enter, we then observe whether they complete their degree by 2013, when the typical student in our sample is about 27 years old. Finally, we observe their labor-market outcomes—formal employment and labor earnings—over 2012 and 2013, when they are 26 to 27 years old, roughly eight to nine years after graduating from high school. Students may enroll in, and graduate from, more than one higher-education program over time; for each student we record the most recently observed (undergraduate) program and use it to define college enrollment, the type of higher education (professional, i.e. academic and university-track, versus technical, i.e. vocational and shorter-cycle), and graduation.[10]

To understand our selection criteria, it is worth noting some institutional aspects of Chile's education system. Primary education consists in grades 1st–8th, and secondary education in grades 9th–12th. Schools, public and private, offer only primary grades, only secondary grades,

[9]These figures follow the students who sat eighth grade in 2000.

[10]Higher-education enrollment is observed from 2005 and degrees from 2007, in both cases through 2013, the last year of the labor-market window, and entry to higher education is restricted to the seven years after this cohort left secondary school in 2004. For 2005 and 2006 the enrollment records come from the centralized college admission system and therefore cover only the universities of the Council of Rectors (CRUCH); the registries we use from 2007 on cover every higher-education institution. A student counts as having entered only if the registry's own first-year-of-enrollment field places her first enrollment in 2011 or earlier. A degree counts for such an entrant, or for a student who began the graduating program by 2011, which also retains those who entered in 2005 or 2006 at institutions outside CRUCH.

or both. Students enrolled in schools offering only primary grades are therefore forced to choose a new school to pursue their secondary education, and for them and their families the period right after completing 8th grade is one of intensive school shopping. This is in contrast with students enrolled in schools offering both levels, for whom the end of 8th grade is not too different from the end of 7th or 9th, and who mostly continue their secondary education in the same school. Since we are interested in students' high school decisions, we restrict our analysis to students enrolled in schools offering only primary education grades.

We then follow Lara et al. (2011) and drop students enrolled in private primary schools, both private-voucher and private-non-voucher. The private-voucher primary sector is by far the larger of the two exclusions, at 83,611 students, or 34.3% of the 243,589 students in the cohort, against 19,989 in private-non-voucher primary schools. Private-voucher primary schools are also much more likely than public ones to continue into secondary grades, so their students do not face the forced choice that defines our design. What remains is arguably the most vulnerable part of the cohort, and the part for which the policy lessons we draw are most relevant. As discussed in Section 2, we also disregard students enrolling in private-non-voucher high schools (Contreras et al., 2010; Lara et al., 2011; Correa et al., 2014); only about 1% of students in public primary schools without secondary grades do so (1,017 of 96,831). We further drop the students for whom no high school is observed, mostly because they never completed secondary school; they represent about 20% of the 121,396 public-primary students who face the forced choice. Finally, we retain observations with missing covariates by including a missing-data category for each covariate, dropping only the small number of students who lack the 8th-grade test scores that constitute our measurement system. Our final estimation sample consists of 94,430 students.

Our data come from the national standardized exams (SIMCE), the censuses of students and schools, the census of students attending higher education institutions, the census of students earning a higher education degree, and the administrative records of Chile's unemployment insurance system, from which we recover formal employment and labor earnings.[11] We use several years for each of these data sets, spanning the period 1998-2016. In addition, we use data from the 2003, 2006, 2009, and 2013 versions of the national household survey (CASEN) to construct the exclusion restrictions we use as part of our identification strategy. These variables relate to the local availability of private-voucher schools, local labor market tightness and opportunities, and local availability of higher education institutions. See appendix A for a more detailed description of each of the data sets we use.

[11]The unemployment insurance database is a matched employer-empoloyee data set that covers the universe of formal firms and workers outside the public sector.

4 Reduced-Form Evidence

We present a first set of results in the form of value-added (VA) regressions, that follow the value-added literature (Rockoff, 2004; Rivkin et al., 2005; Kane and Staiger, 2008; Chetty et al., 2014a,b). We project each labor-market outcome on indicators for the student's final schooling level (or for the type of high school attended), controlling for prior achievement as measured by the four 8th-grade SIMCE standardized test scores. Conditioning on these baseline scores absorbs the prior ability that drives both schooling choices and later earnings, so that the resulting coefficients recover average treatment effects (ATEs)—the mean outcome gain of a schooling level over its baseline, averaged across all students. We estimate these effects on our sample of public-primary 8th-graders who, upon completing primary school, were forced to choose a secondary school. Throughout, the omitted baseline schooling category is a high-school diploma with no (complete or incomplete) higher education.

Concretely, we estimate two specifications. The first relates a labor-market outcome to the student's final schooling level,

$$Y_i = \omega_0 + \omega_1 PG_i + \omega_2 PD_i + \omega_3 TG_i + \omega_4 TD_i + X_i\beta + T_{i,-1}\delta + \varepsilon_i, \quad (1)$$

where Y_i is either formal employment or log annual wage of student i over the 2012-2013 window; PG_i, PD_i, TG_i and TD_i are indicators for the four final schooling levels—professional graduate, professional dropout, technical graduate and technical dropout—with a high-school diploma and no higher education the omitted category; X_i is a vector of demographic covariates (gender, each parent's education, family structure, household income, and region); $T_{i,-1}$ is the vector of student i's four 8th-grade SIMCE scores in verbal, mathematics, social sciences and natural sciences; and ε_i is an idiosyncratic error term. The parameters of interest are ω_1 through ω_4, i.e., the return to each final schooling level relative to a high-school diploma.

The second specification replaces the schooling levels with the high-school sector,

$$Y_i = \gamma_0 + \gamma_1 V_i + X_i\beta + T_{i,-1}\delta + u_i, \quad (2)$$

where V_i takes the value of one if student i attended a private-voucher secondary school and zero if she attended a public one, so that γ_1 is the value-added effect of the private-voucher sector on the labor-market outcome.

We estimate equations (1) and (2) by ordinary least squares, and we cluster standard errors at the 8th-grade school in all regressions. We also estimate equation (1) separately in the voucher and the public subsamples, which lets the returns to higher education differ by the sector in which a student completed secondary school. Under the standard value-added assumption that,

conditional on the covariates and the prior test scores, the remaining variation in schooling is unrelated to the outcome innovation; that is, ω_1 through ω_4 and γ_1 identify average causal effects.

Returns to education. Table 1 reports the VA returns to each final schooling level, relative to a high-school diploma, on formal employment (column 1) and log wages (column 2) by age 28. The wage returns to completion are sizable. Graduating from a professional program raises log wages by 0.405 relative to a high-school graduate, and completing a technical degree raises log wages by 0.239. The striking result is that the returns to higher education are not uniformly positive. Students who enroll in but drop out of a professional program earn 0.263 log points less than they would with only a high-school diploma, whereas technical dropouts earn 0.056 log points more, a small but statistically significant gain. The employment margin orders the same four states differently. Relative to a high-school graduate, professional dropout lowers the probability of formal employment by 8.3 percentage points (p.p.) and professional graduation by 1.9 p.p., while technical graduation leaves it essentially unchanged (-0.3 p.p., statistically indistinguishable from zero) and technical dropout raises it by 3.3 p.p.—consistent with firms valuing the practical skills acquired in vocational programs even absent a completed degree.[12]

Taken together, what higher education pays depends on the type of program as much as on completing it; an incomplete professional spell is the worst labor-market outcome in our cohort on both margins, and an incomplete technical spell is the only state that raises both employment and wages.

[12]Labor income is observed only for the formally employed, through the unemployment-insurance (AFC) administrative records; the wage regressions are therefore estimated on the employed subsample ($N = 72{,}681$), whereas the employment regressions use the full cohort ($N = 94{,}430$). We accordingly report both the extensive (employment) and intensive (wage) margins throughout.

Table 1: Value-Added Returns to Education, Relative to a High-School Diploma

	employment	log wage
	(1)	(2)
professional graduate	-0.019***	0.405***
	(0.005)	(0.018)
technical graduate	-0.003	0.239***
	(0.005)	(0.016)
professional dropout	-0.083***	-0.263***
	(0.005)	(0.019)
technical dropout	0.033***	0.056***
	(0.004)	(0.014)
observations	94,430	72,681

Notes: OLS value-added estimates for our sample of public-primary 8th-graders of 2000. Each regression controls for the four 8th-grade SIMCE test scores; standard errors, in parentheses, are clustered at the 8th-grade school. The omitted category is a high-school diploma with no higher education. Column 1 reports effects on formal employment; column 2, on log wages, estimated on the formally employed subsample. *, **, ***: significant at 10%, 5%, 1%.

The effects of attending a private-voucher high school. We now turn from the type of higher education to the high-school stage itself, and estimate equation (2). Relative to a public-school student with the same baseline achievement, attending a private-voucher high school raises the probability of formal employment by 1.6 percentage points (standard error 0.3) and log wages by 0.073 log points (0.011).

Returns to education by high-school type. Table 2 asks whether the returns to higher education differ by the type of high school a student attended, estimating the returns separately for voucher and public graduates. The branch-specific returns are broadly similar across sectors, but a consistent pattern emerges. Conditional on completing a professional degree, voucher students earn a lower return over a same-sector high-school graduate than public students do (0.369 versus 0.424), while conditional on completing a technical degree they earn a higher one (0.291 versus 0.209); the penalty to professional dropout is also smaller for voucher students (-0.219 versus -0.301). Technical dropouts, by contrast, earn modestly more than a same-sector high-school graduate in both sectors (0.058 versus 0.051, both significant). The employment margin shows similar results. Professional dropout lowers formal employment by 6.9 p.p. for voucher students against 9.2 p.p. for public students, and professional graduation lowers it by 2.4 p.p. for public students but by a statistically insignificant 1.2 p.p. for voucher students; technical dropouts gain about the same in both sectors (3.4 and 3.2 p.p.). The sector differences in the wage returns to a professional degree, to a technical degree, and to professional dropout are modest, between 0.05 and 0.08 log points, but they point in a coherent direction. The private-voucher sector compresses the spread of returns across destinations, adding value in the technical track and

cushioning the downside of an incomplete professional spell, while paying somewhat less for a completed professional degree.

Table 2: Value-Added Returns to Education by High-School Type, Relative to a Same-Sector High-School Diploma

	voucher HS		public HS	
	employment	log wage	employment	log wage
	(1)	(2)	(3)	(4)
professional graduate	-0.012	0.369***	-0.024***	0.424***
	(0.008)	(0.029)	(0.007)	(0.023)
technical graduate	0.010	0.291***	-0.009	0.209***
	(0.009)	(0.026)	(0.006)	(0.020)
professional dropout	-0.069***	-0.219***	-0.092***	-0.301***
	(0.008)	(0.028)	(0.006)	(0.024)
technical dropout	0.034***	0.058***	0.032***	0.051***
	(0.007)	(0.022)	(0.005)	(0.017)
observations	32,844	25,805	61,586	46,876

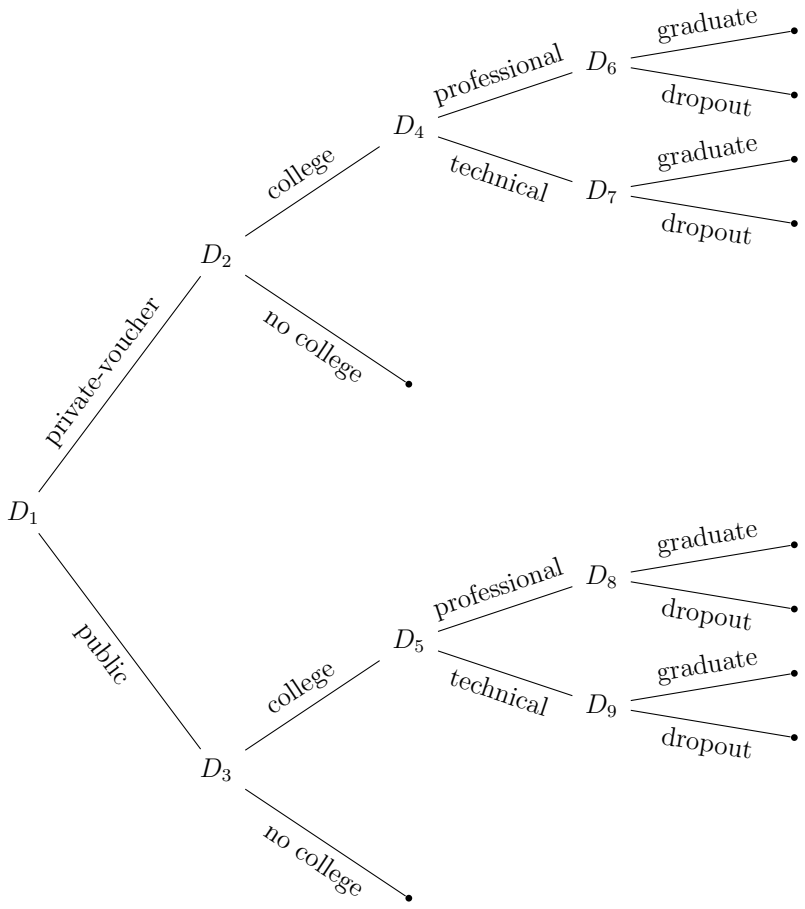
Notes: OLS value-added estimates for our sample of public-primary 8th-graders of 2000, estimated separately for private-voucher and public high-school graduates. Each regression controls for the four 8th-grade SIMCE test scores; standard errors, in parentheses, are clustered at the 8th-grade school. The omitted category within each sector is a high-school diploma with no higher education. Columns 1 and 3 report effects on formal employment; columns 2 and 4, on log wages, estimated on the formally employed subsample. *, **, ***: significant at 10%, 5%, 1%.

The reduced-form evidence just presented establishes the central results of this paper. What higher education pays depends on the type of program as much as on completing it, an incomplete professional spell is the single negative-return outcome on wages, and a private-voucher high school raises earnings directly. What this reduced-form VA approach cannot deliver is a richer set of treatment effect parameters—e.g., which students gain from a professional track and which would have been better served by a technical one, how the returns vary with unobserved ability, and how employment and wages would respond under counterfactual policies that expand or restrict access to the voucher sector. Answering these questions requires modeling the individual schooling decisions that generate them. In the next section, we develop and estimate a model of schooling decisions and labor market outcomes that helps us answer these and other policy-relevant questions.

5 A Sequential Model of Schooling Decisions and Outcomes

We postulate a model of sequential schooling decisions, that starts with the choice of whether to enroll in a private-voucher or public high school, and is followed by all major schooling decisions the student makes thereon, including enrolling in college, choosing between a professional and a technical degree, and obtaining a college degree. Figure 1 displays the complete diagram of the decisions incorporated in our model.

Figure 1: Sequential Schooling Decisions



Notes: This figure displays the multistage decision process we incorporate in our model. Node D_1 relates to the decision of attending a private-voucher or a public high school. Nodes D_2 and D_3 relate to the decision of whether to enroll in college. Nodes D_4 and D_5 relate to the decision of the type of higher education degree to pursue, professional or technical. Nodes D_6–D_9 relate to the decision of whether to graduate from college. A terminal node is denoted by a dot at the end of a branch.

Our model is a generalization of the Roy framework (Roy, 1951; Heckman and Honoré, 1990). We characterize each schooling decision using a flexible discrete choice model, and recognize that individuals make decisions by taking into account the consequences of their choices. In that sense, our model approximates a dynamic discrete choice model without being explicit about the precise rules used by individuals or about their information sets.[13] On the other hand, and similar to reduced-form strategies, we use exclusion restrictions to help pin down individuals' margins of decisions. However, unlike that literature, we are able to define a variety of treatment effect parameters. We also allow for unobserved heterogeneity to play a key role in determining choices and outcomes. Moreover, we are able to interpret the unobserved heterogeneity as a combination of students' inherent scholastic abilities, following the identification argument in Carneiro et al. (2003), and Hansen et al. (2004).[14]

5.1 Schooling Decisions and Counterfactual Outcomes

Let $D_{j'|j}$ denote the decision of attaining schooling level j' given that the student attained schooling level j. For instance, $D_{j'|j}$ may denote the decision of graduating from college given that the student enrolled in college (nodes D_6–D_9 in Figure 1). The optimal decision process is assumed to be characterized by an index threshold-crossing model,

$$D_{j'|j} = \begin{cases} 1 & \text{if } I_{j'|j} \geq 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $I_{j'|j}$ is the perceived value of attaining schooling level j' for a student with schooling level j.

5.2 Parameterization and Assumptions About Unobservables

We use a linear-in-the-parameters model to approximate student's perceived value of transitioning from schooling level j to schooling level j',

$$I_{j'|j} = Z_{j'|j}\gamma_{j'|j} + \theta\lambda_{j'|j} + \nu_{j'|j},$$

where $Z_{j'|j}$ is a vector of contemporaneous observed variables that affect the schooling decision, θ is student's unobserved (to the econometrician) heterogeneity, and $\nu_{j'|j}$ is an idiosyncratic error term. Notice that we do not impose forward-looking behavior or any type of rationality, which

[13] See the discussion in Taber (2000), Cameron and Taber (2004), and Heckman et al. (2018).
[14] See, also, Heckman et al. (2006b), Urzua (2008), Rau et al. (2013), and Rodríguez et al. (2016).

adds flexibility to our model, at the cost of not being able to identify fully structural preference parameters.

Each terminal node of Figure 1, one for each final schooling level under each type of high school, defines a final schooling state s, and we associate with each state two labor-market outcomes, formal employment and log annual earnings, which we parameterize as

$$E_s = \begin{cases} 1 & \text{if } I_s^E \geq 0 \\ 0 & \text{otherwise,} \end{cases} \qquad I_s^E = X^E \beta_s^E + Z_s^E \gamma_s^E + \theta \lambda_s^E + \eta_s^E, \qquad W_s = X^W \beta_s^W + \theta \lambda_s^W + \eta_s^W,$$

where E_s indicates formal employment, W_s is log annual earnings, observed only for the employed, X^E and X^W are vectors of observed variables, Z_s^E is a vector of local labor-market conditions, which enter the employment equation but not the earnings equation, and η_s^E and η_s^W are idiosyncratic error terms. All parameters are state specific, so the model places no restriction on how the returns to schooling differ across final schooling levels or between the two types of high school, and the unobserved heterogeneity θ enters every equation. In what follows we write Y for a generic outcome.

The individual's unobserved heterogeneity plays an important role in our methodology.[15] In particular, we assume that, after controlling for observables, the unobserved heterogeneity drives all statistical dependence between the equations in our model. That is, if the researcher were able to observe θ, she could use matching on (Z, X, θ) to fully identify the model (Carneiro et al., 2003).[16]

Let $\perp\!\!\!\perp$ denote statistical independence. We assume that, conditional on (Z, X),

$$\begin{aligned} \nu_{j'|j} &\perp\!\!\!\perp \nu_{j'''|j''}, && \text{for any } j'|j \neq j'''|j'' \\ \eta_s &\perp\!\!\!\perp \eta_{s'}, && \text{for any } s \neq s' \\ \nu \perp\!\!\!\perp \eta, \quad (\nu, \eta) &\perp\!\!\!\perp \theta, \end{aligned}$$

where ν is the stacked vector of all $\nu_{j'|j}$ terms, $\eta_s = (\eta_s^E, \eta_s^W)$ collects the two outcome errors of terminal state s, and η stacks them across states; within a state we also assume $\eta_s^E \perp\!\!\!\perp \eta_s^W$. In addition, we assume that all unobservables are statistically independent from the observable variables, i.e. $(\nu, \eta, \theta) \perp\!\!\!\perp (Z, X)$.

[15] This is a common feature of the literature on dynamic discrete choice models. See, for example, Keane and Wolpin (1997), and Aguirregabiria and Mira (2010).

[16] (Z, X) denotes the stacked vector of all observables.

5.3 Measurement System for the Unobserved Heterogeneity

We adjoin to our model a measurement system of equations to help identify the distribution of the unobserved heterogeneity, θ, as well as to facilitate its interpretation. The use of a measurement system is not strictly necessary for the model of schooling decisions and outcomes to be identified, and we could instead treat the unobserved heterogeneity as a nuisance or random effect, as is usual in the structural literature (see, e.g., Keane and Wolpin, 1997, and Aguirregabiria and Mira, 2010). However, the use of a measurement system allows us to give a clear interpretation to the unobserved factor as a proxy of the measurements (Carneiro et al., 2003; Hansen et al., 2004).

The measurement system is given by,

$$M_l = X^M \delta_l + \theta \psi_l + \epsilon_l,$$

where M_l is the l-th measurement, X^M is a vector of observed variables determining the measurement, and ϵ_l is an idiosyncratic error term. Notice that the measurements do not depend on the schooling levels, and therefore are observed for all students independent of their schooling choices. In practice, we use measurements taken before the student makes the decisions relevant to our model.

In addition to the independence assumptions mentioned above, we assume that $\epsilon_l \perp\!\!\!\perp \epsilon_{l'} | (Z, X)$ for any $l \neq l'$, and that $\epsilon_l \perp\!\!\!\perp (Z, X, \theta, \nu, \eta)$.

5.4 Identification

The main argument for identification of our model uses a version of matching, and follows Heckman et al. (2016). If θ were observed, then we could condition on (Z, X, θ), and identify all parts of the model. Since θ is not observed, we use the measurement system to proxy θ, and allow for measurement error captured by $\epsilon = (\epsilon_1, \ldots, \epsilon_L)$. Using the conditions presented in Heckman et al. (2016), we can nonparametrically identify our model, including the distributions of θ and ϵ. This last part in the identification requires that we have at least three measurement equations for a one-dimensional factor structure.[17] In practice, we use four measurements.

The factor loadings are identified up to a normalization. We set one of the loadings in the measurement system to be equal to one, which anchors the scale of the factor.

We also benefit from the use of exclusion restrictions, that is, variables that shift the schooling decisions but that do not enter the outcome equations. This identification argument works even without imposing a factor structure in the error terms or invoking the independence conditional on (Z, X, θ) assumption. See Heckman et al. (2016) and the papers cited therein for a formal

[17] See, also, Carneiro et al. (2003), Hansen et al. (2004), and Theorem 1 in Kotlarski (1967).

proof of identification of a general version of a model that shares the characteristics of ours.

5.5 Estimation

We estimate the model by maximum likelihood. The conditional on (Z, X, θ) independence of the error terms assumption is key when constructing the likelihood function, which is given by,

$$\begin{aligned}\mathcal{L} &= \prod_{i=1}^{N} \int f(Y_i, D_i, M_i \mid Z_i, X_i, \theta) f(\theta) d(\theta) \\ &= \prod_{i=1}^{N} \int f(Y_i, D_i \mid Z_i, X_i, \theta) f(M_i \mid X_i, \theta) f(\theta) d(\theta),\end{aligned}$$

where $f(\cdot)$ denotes a probability density function, and we integrate over the distribution of the unobserved factor. We assume mean zero normal distributions for the error terms, with the variances of the threshold-crossing errors $\nu_{j'|j}$ and η_s^E normalized to one and those of η_s^W and ϵ_l estimated freely, and approximate the distribution of the factor using a mixture of two normals. That is,

$$\theta \sim pN(\mu_1, \sigma_1^2) + (1-p)N(\mu_2, \sigma_2^2),$$

where we constrain the factor mean to be zero, and $p \in [0, 1]$.

To ease computation, we perform the estimation in two stages. First, we estimate the measurement system, the factor distribution, and the schooling decisions, using the following first-stage likelihood function,

$$\mathcal{L}^1 = \prod_{i=1}^{N} \int f(D_i \mid Z_i, X_i, \theta) f(M_i \mid X_i, \theta) f(\theta) d(\theta).$$

We obtain estimates $\hat{f}(D_i|Z_i, X_i, \theta)$, $\hat{f}(M_i|X_i, \theta)$, and $\hat{f}(\theta)$, which we use to form the second-stage likelihood function,

$$\mathcal{L}^2 = \prod_{i=1}^{N} \int f(Y_i \mid D_i, Z_i, X_i, \theta) \hat{f}(D_i \mid Z_i, X_i, \theta) \hat{f}(M_i \mid X_i, \theta) \hat{f}(\theta) d(\theta).$$

By proceeding in two stages we reduce the computational burden. Since the schooling decisions, D, and the outcomes, Y, are independent of the measurement system, M, conditional on (Z, X, θ), the two-stage procedure yields consistent estimates.

We use the Gauss-Hermite quadrature for integration, and correct the second-stage standard

errors following the procedure suggested by Murphy and Topel (1985).[18]

6 Empirical Implementation

6.1 Schooling Decisions

As displayed in Figure 1, we model four levels of schooling choices the students make through the course of their education, namely enrolling in a private-voucher high school in lieu of a public high school, enrolling in college, choosing between a professional and a technical degree, and graduating from college (for students enrolled in college). Table 3 displays summary statistics for the schooling choices observed in our sample (students that attended primary public schools offering only primary grades). About a third of students enroll in private-voucher high schools, and about 47% enroll in a higher education institution within seven years of leaving secondary school, 27% in a professional program and 21% in a technical one. Only about 22% of students in our sample earn a higher education degree. Lastly, there is a considerable number of students that drop out from college; of the students ever enrolled in college, only 46% are able to graduate.

6.2 Outcomes

The final outcomes in our model are the individual's formal employment and annual labor earnings in November-2000 US dollars, both measured over 2012 and 2013—when the typical student in our sample is 26 to 27 years old, roughly eight to nine years after graduating from high school—from the unemployment-insurance administrative records described in Section 3. We call this measure a wage throughout, although it is an annual sum over the window in which months without a formal job enter as zero, so it reflects time worked as well as pay. About 77% of the students in our sample hold a formal job at some point over that window. Labor earnings are observed only for the formally employed; we treat the remaining students as not employed, so that employment constitutes the extensive margin of our labor-market outcomes. Table 3 reports summary statistics for these outcomes.

6.3 Measurement System

For the measurement system, we use the 8th grade national standardized exams taken in 2000, which was the first year these tests were administered.[19] The exams are mandatory for all students

[18] See, also, Greene (2008).

[19] In 1999, a similar battery of tests were administered to 4th graders. In subsequent years, different national standardized tests have been administered to students in 2nd, 4th, 6th, 8th, 10th, and 11th grades.

in 8th grade, and evaluate knowledge in four subjects—verbal, mathematics, social sciences, and natural sciences. We include all four tests as part of our measurement system. These tests help us identify the distribution of the unobserved heterogeneity in our model, and allow us to interpret the unobserved factor as a combination of student's inherent scholastic abilities. In the remainder of the paper, we also refer to the unobserved heterogeneity as ability.

We normalize each test score distribution to have mean zero and standard deviation one in the entire population of test takers. Table 3 displays summary statistics for the tests that are part of the measurement system. All four tests have sample means of about -0.14σ, which tell us that on average the students in our sample perform worse than the students in the entire population.

Table 3: Summary Statistics — Endogenous Variables

	mean	std. dev.	min	max
schooling decisions:				
voucher high school	0.35	0.48	0.00	1.00
college enrollment	0.47	0.50	0.00	1.00
professional enrollment	0.27	0.44	0.00	1.00
technical enrollment	0.21	0.40	0.00	1.00
college graduation (cond. enrolled)	0.46	0.50	0.00	1.00
college degree (unconditional)	0.22	0.41	0.00	1.00
labor market:				
employed	0.77	0.42	0.00	1.00
log annual income (if employed)	7.93	1.27	-3.00	10.55
measurement system:				
verbal	-0.14	0.91	-2.81	2.90
math	-0.14	0.91	-2.75	2.63
social sciences	-0.15	0.91	-2.84	3.06
natural sciences	-0.14	0.92	-2.76	2.81

Notes: Summary statistics for the endogenous variables in our estimation sample (N=94,430). Professional and technical enrollment are unconditional shares (they sum to college enrollment); college graduation is conditional on having enrolled, and college degree is unconditional. Labor income is observed only for the formally employed (the income row is computed on that subsample); log annual income is the log of earnings over the 2012–2013 unemployment-insurance window, in November-2000 US dollars. The 8th-grade (measurement-system) test scores are normalized to mean zero and standard deviation one in the population of test takers.

6.4 Covariates

The exogenous variables we use in our model's equations mostly reflect socio-economic characteristics, and are common in the related education/labor literature (see, e.g., Altonji, 1993; Heckman et al., 2006b; Walters, 2017; Heckman et al., 2018). Table 4 lists the covariates we use, and displays summary statistics for each of them. The variables for gender, parents' education,

broken home, and household income are measured when the student is 14 years old; we also observe region of residence at ages 14 and 18. The variable for broken home is a dummy that takes the value of one if no parent or only one of them resides in the student's home, and zero otherwise. Because these socio-economic variables are missing for a non-trivial share of students, we enter parents' education, broken home, and household income as categorical indicators with a separate missing-data category to avoid dropping the affected observations. About half of our sample is male. Reflecting the public-primary population we study, most parents did not complete secondary education; only about a quarter of mothers and fathers have twelve or more years of schooling, and roughly a fifth have missing parental education. Household income is reported in brackets, with about 29% of households earning between CLP 100K and CLP 200K per month and 17% earning more than CLP 200K. Lastly, about a sixth of the students in our sample reside in the country's Northern region, and about half reside in the Central region, with the rest in the South.

6.5 Exclusion Restrictions

As stated in Section 5.4, our model is nonparametrically identified without the need of exclusion restrictions, and invoking a form of conditional independence assumption on mismeasured variables. However, we could also identify all parameters in our model with the use of node-specific instruments, or variables that shift the schooling choices but that do not enter the outcome equations. With instruments, identification is secured even without relying on the matching-like assumption just mentioned.

We use a variety of instruments that shift the schooling decisions. Table 4 lists them, and provide summary statistics for each of them. The variable for the percentage of private-voucher high schools in the municipality is measured when the student is 14 years old. The variables for local unemployment and wages that shift the college enrollment and graduation decisions are differences between high-skilled and low-skilled municipality averages. The 2013 variables, which enter the employment equations but not the earnings equations, are instead the municipality levels, the unemployment rate and the log of mean monthly earnings among residents aged 18 to 35, computed among residents holding a higher-education degree for the graduate states and among those without one for the remaining states. The variable for presence of a college in the municipality is a dummy that is equal to one if there is one or more colleges in the student's municipality of residence, and zero otherwise. On average, students live in a municipality where 53% of high schools are private-voucher. The local labor-market differentials capture the relative tightness of the skilled labor market; high-skill unemployment rates are on average about 4 percentage points lower than low-skill rates, and high-skill wages exceed low-skill wages. About

60% of students reside in a municipality with at least one college, and about 52% in one with a professional college. Notice that we observe many of the instruments at different moments in the course of the student's education.

Table 4: Summary Statistics — Exogenous Variables

	mean	std. dev.	min	max
covariates:				
male	0.51	0.50	0.00	1.00
mother's education: 12 years	0.19	0.39	0.00	1.00
mother's education: more than 12 years	0.06	0.23	0.00	1.00
mother's education: missing	0.17	0.38	0.00	1.00
father's education: 12 years	0.21	0.41	0.00	1.00
father's education: more than 12 years	0.07	0.26	0.00	1.00
father's education: missing	0.20	0.40	0.00	1.00
broken home	0.23	0.42	0.00	1.00
broken home: missing	0.23	0.42	0.00	1.00
household income: CLP 100–200K	0.29	0.46	0.00	1.00
household income: more than CLP 200K	0.17	0.37	0.00	1.00
household income: missing	0.25	0.43	0.00	1.00
region: north (age 14)	0.16	0.37	0.00	1.00
region: center (age 14)	0.51	0.50	0.00	1.00
region: north (high school)	0.16	0.37	0.00	1.00
region: center (high school)	0.51	0.50	0.00	1.00
exclusion restrictions:				
% voucher schools in municipality	0.53	0.25	0.00	1.00
local unemployment gap (high–low skill, 2003)	-0.04	0.05	-0.21	0.32
local wage gap (high–low skill, 2003)	4.56	2.67	0.03	33.59
college in municipality	0.60	0.49	0.00	1.00
professional college in municipality	0.52	0.50	0.00	1.00
local unemployment gap (high–low skill, 2009)	-0.04	0.05	-0.22	0.26
local wage gap (high–low skill, 2009)	4.46	2.67	-0.86	42.18
local unemployment, high-skill (2013)	0.09	0.10	0.00	1.00
local unemployment, low-skill (2013)	0.13	0.06	0.00	0.69
local wage, high-skill (2013)	6.39	0.30	5.24	7.71
local wage, low-skill (2013)	5.74	0.20	4.74	6.52

Notes: Summary statistics for the covariates and exclusion restrictions in our model (see Table B.1 for their allocation across equations). Education, broken-home, household-income, and region variables enter as categorical indicators; the omitted categories are education below 12 years, non-broken home, household income below CLP 100K per month, and the southern region, with a separate missing-data category for each. Variables measured at age 14 (the primary-school stage) enter the measurement system and the high-school choice; their high-school-stage analogues enter the later equations. The exclusion restrictions are the local availability of private-voucher schools, of a college, and of a professional college, and the local labor-market variables measured at different ages, the high–low skill gaps that shift the schooling decisions and the 2013 skill-group levels that shift employment.

Finally, Table B.1 in Appendix B displays the exact inclusion of the covariates and instruments in the various equations of our model.

7 Results

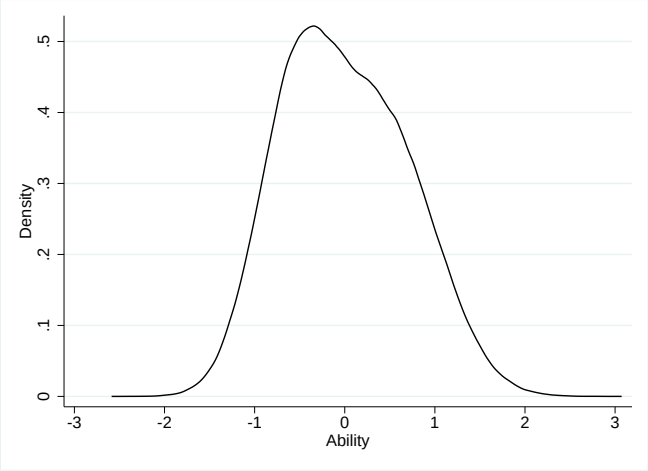
Our results proceed in three steps. We first ask what each schooling option is worth to a student who is choosing among them. We then ask what attending a private-voucher high school does to the schooling options students actually take up. Finally, we put the two together, decomposing the effect of vouchers on labor market outcomes into the schooling flows they induce and the returns those destinations pay. We begin with the estimates and the fit of the model.

7.1 Estimates

7.1.1 Distribution of the Unobserved Heterogeneity

Figure 2 displays the estimates for the distribution of the unobserved heterogeneity. It also plots the estimated distribution. A first look at the plotted distribution suggests its departure from the normal distribution. The two components of the mixture are centered on opposite sides of zero, one at 0.42 and the other at -0.55, and the estimated density is correspondingly asymmetric, with its mode on the negative side and a shoulder on the positive side. This result confirms our approach of allowing for a degree of flexibility in the distribution of the factor.

Figure 2: Distribution of Ability



$$f \sim pN(\mu_1, \sigma_1^2) + (1-p)N(\mu_2, \sigma_2^2)$$

where

$$(\mu_1, \mu_2) = (0.42, -0.55)$$
$$(\sigma_1, \sigma_2) = (0.58, 0.43)$$
$$p = 0.56$$

Notes: This figure displays the estimated distribution of the unobserved factor. Estimated parameters were obtained by maximum likelihood in the first stage, jointly with the measurement system and the schooling decisions.

7.1.2 Measurement System

The estimates for the measurement system are reported in Table B.2 in Appendix B. Consistent with other studies for the case of Chile, we find that female students outperform males in verbal, but the opposite occurs in math (Rau et al., 2013; Rodríguez et al., 2016). A gender effect also exists in social sciences, favoring males. Both mother's and father's education are strong determinant of students' performance, although the effect of mother's education is estimated to be stronger in all subjects. Residing in a household without both parents being present has a negligible effect on test scores. That is not the case for household income, which increases students' performance. Residing in the Southern region also increases test scores. Finally, the unobserved ability factor is estimated to be a strong determinant of test scores, playing a comparable role in all subjects.

7.1.3 Schooling Decisions

Estimates for the decision of attending a private-voucher high school in lieu of a public high school are presented in Table B.3 in Appendix B. This decision corresponds to node D_1 in Figure 1. Males are more likely to enroll in a voucher high school. Higher levels of mother's education increases the likelihood that a student attends a private-voucher high school. The effect of father's education is not statistically different from zero. Higher levels of household income also increase the likelihood of attending a voucher school. Residing in the Northern region decreases the chances of going to a private school, and the opposite occurs with residency in the Central region; all, relative to residing in the South. Notice that the exclusion restriction, which captures local availability of voucher high schools, is a strong determinant of the decision. Lastly, higher levels of the unobserved ability increase the likelihood of enrolling in a voucher school.

Table B.3 also reports the estimates for the decision of enrolling in college, which corresponds to the nodes D_2 and D_3 in Figure 1, for students coming from private-voucher and public high schools, respectively. Female students are more likely to enroll in college. Both mother's and father's education strongly increase the likelihood of enrolling, with the effect rising in the level of parental schooling. Not having both parents at home reduces the chances of enrolling. Higher levels of household income substantially increase enrollment. Among public high school students, residing in the Northern or Central regions lowers enrollment relative to the South, whereas the regional effects are negligible for voucher students. We use three exclusion restrictions in these equations. The first is the difference between the local unemployment rate of high-skilled and low-skilled workers; the corresponding coefficients are negative in both equations, statistically significant for voucher students but not for public ones. The second is the difference between the local log wage of high-skilled and low-skilled workers, whose estimated effect is close to zero. Our third instrument is the local availability of a higher education institution, whose coefficient is positive and significant, indicating that the presence of a college in the student's municipality of residence increases the likelihood of enrolling. Lastly, the loading for the unobserved ability factor is strongly positive, meaning that higher levels of ability markedly increase the likelihood of enrolling in college.

Estimates for the decision of pursuing a professional (instead of a technical) college degree are presented in Table B.4 in Appendix B. This decision corresponds to the nodes D_4 and D_5 in Figure 1, for students who enrolled in college coming from private-voucher and public high schools, respectively; the dependent variable equals one for a professional degree and zero for a technical degree. Female students are more likely to pursue a professional degree. Family endowments matter. Higher levels of parents' education and household income increase the likelihood of choosing a professional over a technical degree, while growing up without both parents present reduces it. As an exclusion restriction in these equations we use the local availability of a profes-

sional higher education institution, whose coefficient is positive (and statistically significant for public high school students), indicating that the local presence of a professional college raises the probability of pursuing a professional degree. Lastly, the loading for the unobserved ability factor is strongly positive, meaning that higher-ability students are considerably more likely to opt for a professional degree.

The decision of graduating from college is estimated only for students who enroll in college, and its estimates are presented in Table B.5 in Appendix B. This decision corresponds to the nodes D_6–D_9 in Figure 1, which distinguish students by high school type (private-voucher or public) and by the type of degree pursued (professional or technical). Female students are more likely than males to graduate from college across all four paths. Mother's education is generally a positive determinant of graduation, whereas the effect of father's education is smaller and less consistent. Growing up in a household without both parents present decreases the chances of obtaining a degree, while higher household income increases them, particularly for students from private-voucher high schools. As exclusion restrictions we use the local labor market differentials in unemployment and wages between high-skilled and low-skilled workers, whose coefficients are mostly imprecisely estimated. Lastly, the loadings for the unobserved ability factor are positive throughout, indicating that higher-ability students are more likely to earn a college degree. This is, again, in line with the related evidence (Heckman et al., 2006b, 2018; Rau et al., 2013; Rodríguez et al., 2016).

7.1.4 Labor-Market Outcomes

Estimates for the probability of holding a formal job are presented in Table B.6 in Appendix B. We estimate one employment equation for each of the ten terminal schooling states, that is, for each of the five final schooling levels under each type of high school. Males are more likely to hold a formal job in every state, and their advantage is smallest among the students who entered a professional program and considerably larger among technical students and among those who never enrolled. Residing in the Central region raises the probability of formal employment in almost every state, whereas residing in the Northern region raises it in three of the five public states and has no systematic effect among voucher ones; all, relative to residing in the South. The local labor market enters these equations through the unemployment rate and the wage of the student's own skill group, which shift employment but not wages. A higher local unemployment rate reduces the probability of formal employment, and a higher local wage raises it, in nine of the ten states in each case, although both are imprecisely estimated in several of them. Lastly, the loadings for the unobserved ability factor are small, and they change sign across schooling states. Higher ability lowers the probability of formal employment among students who entered a professional program, whether or not they graduated, and slightly raises it among technical

students and among those who never enrolled.

Estimates for log annual wages are presented in Table B.7 in Appendix B. Here too we estimate one equation per terminal schooling state, on the formally employed. Males earn more than females in every state, and the gap is at its narrowest among professional graduates and at its widest among the students who never enrolled in higher education. Students residing in the Northern and Central regions earn more than those residing in the South in nearly every state. Under both types of high school the professional degree carries the highest constant, followed by the technical degree, the incomplete technical spell and the high-school diploma, while the incomplete professional spell carries the lowest, below the diploma. The estimated dispersion of wages follows a different ranking. It is largest in the two professional states under both types of high school, and largest of all in the incomplete professional spell, whereas the technical states and the high-school diploma are considerably less dispersed, which anticipates an asymmetry in risk between the two tracks that we later explore. Lastly, the loadings for the unobserved ability factor are positive in every state except the two incomplete professional spells, where they are indistinguishable from zero, and they are small relative to the residual standard deviations, so the unobserved factor accounts for little of the dispersion of wages within a schooling state.

7.2 Goodness of Fit

To test the ability of our empirical model to reproduce the actual data, we simulate 500,000 observations using the model, its estimates, and the data. The exercise is as follows. We randomly select an observation from the data, and draw a value for each unboservable in the model from their corresponding estimated distributions. We then compute the predicted value of the indirect utility in every decision node of the model, which allows to obtain the simulated optimal choice in each of those nodes, by selecting the decision with the highest indirect utility value. We also predict counterfactual outcomes (i.e., employment and wages) associated with every final schooling state. We repeat this process 500,000 times.

Tables B.8–B.10 in Appendix B compare our model's predictions with the data. Table B.8 reports comparisons for the measurement system, Table B.9 for schooling decisions, and Table B.10 for formal employment and log wages, within each of the ten terminal schooling states and across quartiles of the 8th-grade test scores, which enter the outcome equations only through the unobserved factor. In general, our model fits the data satisfactorily well, which gives us confidence to use the model to compute counterfactuals, and more generally to derive answers to policy-relevant questions.

7.3 The Returns to Education, and the Risk of Enrolling

7.3.1 Returns to Education

We begin our examination of the returns to education for our population of public-primary students by comparing the labor market outcomes associated with different final schooling levels, in a pairwise fashion. To facilitate exposition, we contrast each final schooling level with respect to high school diploma. Specifically, we compute,

$$\begin{aligned} ATE_{j,0} \;=\; & \int\int E(Y_j - Y_0 \mid X = x, \theta = \bar{\theta}, D_1 = 1) \Pr(D_1 = 1 \mid X = x, \theta = \bar{\theta}) + \\ & E(Y_j - Y_0 \mid X = x, \theta = \bar{\theta}, D_1 = 0) \Pr(D_1 = 0 \mid X = x, \theta = \bar{\theta}) dF_{X,\theta}(x, \bar{\theta}), \end{aligned}$$

which defines the average treatment effect of attaining final schooling level j relative to obtaining a high school diploma (denoted by the subscript 0), and were we weight each individual's treatment effect by her likelihood of attending a private-voucher and a public high school. The set of final schooling levels we analyze includes professional graduate, technical graduate, professional dropout, technical dropout. This exercise is akin to what is obtained from a linear regression of outcomes projected onto final schooling levels dummies, with high school diploma the base category (Rodríguez et al., 2016).

We compute our estimates for the ATE parameters by averaging over the $Y_j - Y_0$ distribution, using our simulations from the model. Table 5 displays the results. Panel A shows returns to education on employment, while panel B does similarly on log annual wages. In addition to the average effect in the overall distribution, we explore heterogeneity by reporting returns separately for students in the bottom, middle, and top tercile of the ability distribution, in columns (2) to (4). All final schooling levels, except for technical dropout, are associated with reducing formal employment. This finding is possibly explained by high school graduates entering the labor market sooner than individuals with some higher education, having accumulated more work experience and on-the-job knowledge by age 27. Graduating from a professional program reduces formal employment by 4.9 p.p., whereas graduating from a technical one reduces it by only 1.5 p.p. Dropping out from a professional degree doubles the professional graduate's employment reduction, to 9.8 p.p., while technical degree dropout is associated with increasing formal employment by 2.2 p.p., possibly reflecting the value that firms put on the practical skills acquired at technical schools, on top of the ability signal of the degree (Arteaga, 2018).

The heterogeneity is concentrated in the dropout states. The employment penalty to professional dropout deepens monotonically with ability, from 6.4 p.p. in the bottom tercile to 13.6 p.p. in the top, whereas the penalty to a professional degree steepens less sharply, from 3.1 p.p. at the bottom to 6.9 p.p. at the top. Technical dropout, by contrast, raises employment throughout,

and by more for higher-ability students.

Panel B shows that higher education degrees are associated with substantial returns to education. Obtaining a professional degree increases wages by 0.36 log points relative to a high school diploma, while a technical degree increases wages by 0.24 log points. Interestingly, the wage returns to dropping out of a professional degree are negative and large, at -0.23 log points. Dropping out of a technical program, by contrast, raises wages by a small but precisely estimated 0.06 log points. The negative return to an incomplete professional spell contrasts with evidence for the United States, where some college without a degree still tends to carry a positive wage return (Heckman et al., 2018). Read together, the two panels show that what higher education pays depends on the type of program as much as on completing it; an incomplete professional spell is the only destination whose wage return is negative, and it is also the one that costs most on the employment margin. The penalty is also strongly graded by ability, rising from -0.11 for students in the bottom tercile to -0.37 for those in the top, so the students likeliest to complete a professional degree are also those who lose most by not finishing it. The return to a professional degree, by contrast, is essentially the same at every ability level, at about 0.36 in each of the three terciles.[20]

[20]Interestingly, the returns do not vary with family background. Table B.11 in Appendix B splits the sample by mother's education (less than high school, and at least high school), and the returns are essentially flat across the two groups; the wage return to a professional degree is 0.37 for students with mother's with no high school diploma and 0.36 for their peers with mothers with at least high school, and the remaining schooling levels are similarly close.

Table 5: Returns to Education for Final Schooling Levels

		ability tercile		
	all	low	middle	high
	(1)	(2)	(3)	(4)
A. Employment				
professional graduate	-0.049***	-0.031***	-0.049***	-0.069***
	(0.006)	(0.009)	(0.006)	(0.006)
technical graduate	-0.015***	-0.021***	-0.015***	-0.008
	(0.005)	(0.008)	(0.005)	(0.007)
professional dropout	-0.098***	-0.064***	-0.094***	-0.136***
	(0.005)	(0.007)	(0.005)	(0.006)
technical dropout	0.022***	0.016**	0.021***	0.030***
	(0.004)	(0.006)	(0.004)	(0.007)
B. Log wages				
professional graduate	0.363***	0.357***	0.366***	0.366***
	(0.024)	(0.034)	(0.024)	(0.025)
technical graduate	0.235***	0.209***	0.241***	0.255***
	(0.020)	(0.029)	(0.020)	(0.027)
professional dropout	-0.231***	-0.107***	-0.218***	-0.367***
	(0.023)	(0.034)	(0.023)	(0.026)
technical dropout	0.061***	0.085***	0.068***	0.030
	(0.018)	(0.023)	(0.018)	(0.026)

Notes: Estimated average treatment effects (ATE) of each final schooling level relative to a high school diploma, on employment (Panel A) and log annual wages (Panel B). Columns (2) to (4) split students into terciles of the ability distribution. Point estimates are from the model; standard errors in parentheses are from a parametric bootstrap (500 draws of the parameters from their estimated asymptotic distribution). *, **, ***: significant at 10%, 5%, 1%.

It is reassuring that these structural returns line up closely with the reduced-form value-added estimates of Section 4, which rest on a different identifying assumption. The structural wage return to a professional degree, 0.36, is within roughly four log points of the value-added estimate of 0.41; the structural penalty to professional dropout, -0.23, is within three log points of the value-added -0.26; and the technical-degree return is essentially the same under both methods, 0.24. That two methods resting on different assumptions deliver nearly the same average wage returns lends credibility to the structural model, which we now use to isolate the role of vouchers.

7.3.2 Returns to Enrolling

We now investigate the returns to enrolling in either a professional or a technical track. We focus both on the expected gains as well as on the risk that comes attached to enrolling. We compare the outcome distributions of the options, and the differences between their quantiles. Table 6 reports the results. Enrolling in a professional program raises the expected log wage by a statistically insignificant 0.019 relative to stopping at a high school diploma, while enrolling in a technical

program raises it by 0.140. The 0.122 log point gap between the two (Panel A, technical minus professional) is precisely estimated. On employment the ordering is starker still. The professional option lowers the probability of formal employment by 7.7 p.p., while the technical option leaves it statistically unchanged. Professional programs do pay more in wages when completed—the 0.36 log point degree premium of Table 5—but fewer than two in five students would finish one (Panel C), and an incomplete professional spell is worse than never having enrolled on both margins.

The professional track is also the riskier of the two. Panel B measures that risk in the form of the variance of log wages, and performs a variance decomposition exercise. We take the log wage a student faces at the moment she chooses an option, before she knows whether she will complete the program and before her own wage is drawn, and we split its variance into three terms that add up to the total. The first is the completion lottery, the spread created by landing in the graduate relative to the dropout state, which depends on the student's own chance of finishing and on how far apart the returns to the two states are. The second is the dispersion of wages within whichever state she reaches. The third is the dispersion due to differences in students' ability and background.

The variance of the log wage is 2.05 under the professional option, against 1.41 under the technical one, and 1.49 for a high school diploma. The decomposition of those variances is as follows. The dispersion of wages within terminal states accounts for most of the variance of every option, and it is what sets the professional option apart, at 1.93 against 1.31 for the technical option, and 1.35 for a high school diploma, largely because wages are so dispersed among professional dropouts. Differences in ability and background contribute least to the professional option, 0.03 against 0.09 for the technical one, and 0.15 for a high school diploma. Completion risk, the share of the variance due to the completion lottery, is a small part of the total: 4.4% for the professional option and 0.6% for the technical one. The gap comes almost entirely from the much wider distance between the graduate and dropout outcomes on the professional track, as opposed to from the difference in the chance of finishing.

Table 6: The Returns to Enrolling: Professional versus Technical

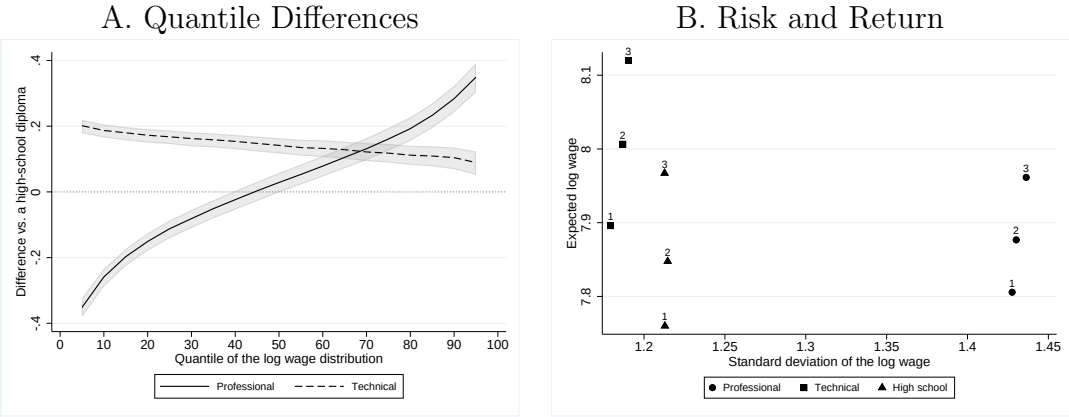
	professional (1)	technical (2)	high school (3)
A. What the option delivers			
expected log wage	7.879	8.001	7.860
	(0.016)	(0.012)	(0.008)
return to enrolling	0.019	0.140***	—
	(0.018)	(0.014)	
technical minus professional	—	0.122***	—
		(0.020)	
probability of employment	0.714	0.796	0.791
	(0.003)	(0.003)	(0.002)
return to enrolling	-0.077***	0.005	—
	(0.004)	(0.003)	
B. Risk			
variance of log wage	2.051	1.410	1.494
	(0.021)	(0.013)	(0.008)
completion lottery	0.091	0.009	—
	(0.008)	(0.002)	
wages within terminal states	1.926	1.310	1.347
	(0.019)	(0.012)	(0.007)
ability and background	0.033	0.092	0.147
	(0.004)	(0.006)	(0.005)
completion risk (percent of variance)	4.4	0.6	—
	(0.4)	(0.1)	
C. The gamble			
probability of graduating	0.390	0.438	—
	(0.004)	(0.004)	

Notes: Each column describes the outcome distribution a student faces upon choosing that schooling option, integrating over her own subsequent graduation decision. Column (3) is the high-school-diploma baseline, so the rows labeled *return to enrolling* are columns (1) and (2) minus column (3). Expected log wage and the variance components are computed from the model's conditional means and completion probabilities given ability and covariates; the variance of an option decomposes into the completion lottery, the dispersion of wages within a terminal state, and the dispersion of conditional means across students, and the three add up to the variance reported in the first row of Panel B, up to rounding. The technical-minus-professional row is the difference between the two returns to enrolling. The probability of graduating is the share of all students who would complete that program if they enrolled, and not the completion rate among those who do enroll, which is higher on both tracks because students who choose a program are selected on what predicts completion. Standard errors in parentheses are from a parametric bootstrap (500 draws of the parameters from their estimated asymptotic distribution). *, **, ***: significant at 10%, 5%, 1%.

Figure 3 makes the comparison concrete. Panel A plots, at each quantile, the difference between the option's log wage distribution and the high-school baseline. The technical option shifts the distribution up at every quantile, by between 0.09 and 0.20 log points, and by more at the bottom of the distribution than at the top. The professional option instead rotates it.

At the fifth percentile it pays 0.35 log points less than a high school diploma, it crosses zero at about the forty-fifth percentile, and by the ninety-fifth it pays 0.35 more. The confidence intervals show that the rotation is precisely estimated, with the professional curve significantly negative at and below the thirty-fifth percentile and significantly positive from the fifty-fifth. Enrolling in a professional program represents a spread bet, while enrolling in a technical one is close to a sure gain. Panel B places the two options in the mean–standard-deviation plane, by ability tercile, and shows the dominance of the technical track directly, since every professional point lies below and to the right of its technical counterpart.

Figure 3: The Distribution of Returns to Enrolling



Notes: Panel A plots, at each quantile of the log wage distribution, the difference between the distribution a student faces upon enrolling in a professional (solid) or technical (dashed) program and the distribution she faces with a high school diploma. Shaded areas are pointwise 90% parametric-bootstrap bands over the 500 parameter draws. Panel B plots the expected log wage of each option against its standard deviation, separately by ability tercile (labeled 1, 2, 3). Both panels are computed from the model at the estimated parameters.

Table 7 complements the analysis by examining the returns to enrolling in a professional or a technical program by ability tercile. The expected log wage under the professional option rises from 7.81 in the bottom ability tercile to 7.96 in the top, but the technical option rises faster, from 7.88 to 8.13, so the gap widens with ability, from 0.077 log points at the bottom to 0.163 at the top. The employment penalty to the professional option also widens, from 5.2 p.p. at the bottom to 10.3 p.p. at the top. Indeed, for students in the top tercile the professional option delivers an expected log wage (7.96) no higher than stopping at a high school diploma (7.98), whereas the technical option delivers 0.152 log points more. Higher-ability students are likelier to finish—the share of students who would complete a professional program if they enrolled rises from 0.32 in the bottom tercile to 0.47 in the top—but the wage penalty they pay when they do not finish deepens faster than their completion rises, so the returns to enrolling in a professional

program fall with ability, from 0.055 at the bottom to a statistically insignificant −0.011 at the top. Their forgone employment is also larger.

Table 7: The Returns to Enrolling, by Ability

	low	middle	high
	(1)	(2)	(3)
A. Enroll in a professional program			
expected log wage	7.807	7.867	7.964
	(0.025)	(0.016)	(0.015)
return to enrolling	0.055**	0.012	-0.011
	(0.027)	(0.018)	(0.020)
probability of employment	0.735	0.715	0.692
	(0.006)	(0.003)	(0.003)
probability of graduating	0.319	0.385	0.467
	(0.006)	(0.004)	(0.004)
B. Enroll in a technical program			
expected log wage	7.883	7.992	8.127
	(0.017)	(0.012)	(0.017)
return to enrolling	0.132***	0.137***	0.152***
	(0.020)	(0.014)	(0.021)
probability of employment	0.788	0.796	0.804
	(0.004)	(0.003)	(0.005)
probability of graduating	0.364	0.433	0.518
	(0.005)	(0.004)	(0.006)
C. Stop at a high-school diploma			
expected log wage	7.752	7.855	7.975
	(0.009)	(0.008)	(0.013)
probability of employment	0.787	0.791	0.795
	(0.002)	(0.002)	(0.003)
D. Technical minus professional			
difference in expected log wage	0.077**	0.125***	0.163***
	(0.031)	(0.020)	(0.023)
difference in employment	0.053***	0.081***	0.111***
	(0.007)	(0.004)	(0.006)

Notes: Columns split students into terciles of the ability distribution. Each panel describes the outcome distribution a student faces upon choosing that option, integrating over her own subsequent graduation decision; *return to enrolling* is measured relative to the high-school-diploma baseline in Panel C. Standard errors in parentheses are from a parametric bootstrap (500 draws); stars are reported only for the rows that are differences, since a test against zero is not informative for a level or a probability. *, **, ***: significant at 10%, 5%, 1%.

7.4 The Effects of Vouchers

We now turn our attention to what attending a private-voucher high school in lieu of a public one does to the schooling students end up with and to their labor market outcomes. Since the

model delivers each student's counterfactual outcomes under both branches, we can compute the relative gain from attending a private-voucher school, $Y_1 - Y_0$, for every simulated individual. Averaging those gains over all students defines the average treatment effect:

$$ATE \quad = \quad \int \int E(Y_1 - Y_0 \mid X = x, \theta = \bar{\theta}) dF_{X,\theta}(x, \bar{\theta}),$$

which is the gain a randomly chosen student would obtain by attending a private-voucher high school instead of public one. In this section, we focus on this treatment effect parameter.

We begin by investigating whether the returns to education presented in Section 7.3 depend on the sector in which a student completed secondary school. Table 8 reports the returns to each final schooling level a randomly chosen student would earn were all students forced onto the voucher branch, in column (2), and onto the public branch, in column (3). Two patterns emerge. The wage return to a professional degree is lower under the voucher branch (0.315) than under the public one (0.393), while the return to a technical degree is higher (0.265 versus 0.216). The incomplete spells, by contrast, pay similarly across sectors; an incomplete technical spell is worth 0.060 log points under the voucher branch and 0.066 under the public one, significantly positive in both, and an incomplete professional spell costs 0.232 under the voucher branch and 0.242 under the public one. Among degrees, then, the private-voucher sector raises what a technical degree pays and lowers what a professional one pays—consistent with our reduced-form value-added estimates in Table 2.

Table 8: Returns to Education by High-School Type

	all	voucher	public
	(1)	(2)	(3)
A. Employment			
professional graduate	-0.049***	-0.039***	-0.055***
	(0.006)	(0.008)	(0.007)
technical graduate	-0.015***	-0.010	-0.019***
	(0.005)	(0.008)	(0.006)
professional dropout	-0.098***	-0.091***	-0.106***
	(0.005)	(0.007)	(0.006)
technical dropout	0.022***	0.025***	0.021***
	(0.004)	(0.008)	(0.005)
B. Log wages			
professional graduate	0.363***	0.315***	0.393***
	(0.024)	(0.037)	(0.031)
technical graduate	0.235***	0.265***	0.216***
	(0.020)	(0.035)	(0.026)
professional dropout	-0.231***	-0.232***	-0.242***
	(0.023)	(0.034)	(0.030)
technical dropout	0.061***	0.060**	0.066***
	(0.018)	(0.029)	(0.023)

Notes: Estimated returns to each final schooling level relative to a high school diploma, on employment (Panel A) and log annual wages (Panel B). Columns (2) and (3) report the average treatment effect for a randomly chosen student were all students forced to attend a voucher (2) or public (3) high school, computed from the model's counterfactual outcomes for the whole population. Column (1) is the overall ATE. Standard errors in parentheses are from a parametric bootstrap (500 draws). *, **, ***: significant at 10%, 5%, 1%.

Table 9 proceeds by reporting the effects of private-voucher attendance on every schooling margin the model distinguishes—enrollment overall and by track in Panel A, the five terminal schooling states in Panel B, and completion in Panel C. Column (1) gives the share under the public branch, so each effect can be read against its own base, and column (2) reports the average treatment effect. Standard errors throughout are from a parametric bootstrap over the estimated parameters.

Attending a private-voucher high school raises higher-education enrollment by 1.97 p.p. on average, against a public-branch enrollment rate of 46.7%. The effect is precisely estimated, and its magnitude is in line with the small existing literature on vouchers and college enrollment (Chingos and Peterson, 2015; Chingos, 2018). Importantly, the entire increase is professional. Professional enrollment rises by 2.01 p.p., while technical enrollment does not move (-0.05 p.p., statistically zero). In other wors, private-voucher schools expand higher education exclusively along the track that Section 7.3 showed to be dominated.

Panel B displays the final schooling choices private-voucher schools induce students toward. The 1.97 p.p. pulled out of non-enrollment end up mostly as incomplete spells. Professional

dropout rises by 1.37 p.p., more than twice the 0.64 p.p. rise in professional graduation, and technical graduation falls by 0.77 p.p., more than offsetting the professional degrees gained. Panel C complements these figures. Attending a private-voucher high school raises the probability of having started higher education and not finished it by 2.09 p.p., and leaves the probability of holding any higher-education degree no higher (−0.13 p.p., statistically zero).

Table 9: The Effects of a Private-Voucher High School on Schooling Choices

	public base (1)	ATE (2)
A. Enrollment		
any higher education	46.71	1.97***
		(0.32)
professional	25.92	2.01***
		(0.29)
technical	20.78	-0.05
		(0.30)
B. Terminal schooling state		
not enrolled	53.29	-1.97***
		(0.32)
professional graduate	12.11	0.64***
		(0.22)
professional dropout	13.82	1.37***
		(0.24)
technical graduate	9.59	-0.77***
		(0.21)
technical dropout	11.20	0.72***
		(0.23)
C. Completion		
any degree	21.70	-0.13
		(0.28)
any dropout	25.01	2.09***
		(0.30)

Notes: Average treatment effects of attending a private-voucher rather than a public high school on schooling choices, in percentage points. Column (1) is the share under the public branch. Panels A–C are shares of all students, so professional and technical enrollment sum to any higher education, and any degree and any dropout sum to it as well. Standard errors in parentheses are from a parametric bootstrap (500 draws). *, **, ***: significant at 10%, 5%, 1%.

Table 10 reports the voucher-induced change in the share of students reaching each terminal state within terciles of the ability distribution. Panel A shows that non-enrollment falls by 2.79 p.p. in the bottom tercile against 1.30 p.p. in the top, so more than three-quarters of the students vouchers bring into higher education come from the bottom two terciles. In the bottom tercile the inflow splits 1.92 p.p. into professional dropout and 1.14 p.p. into technical dropout against 0.75

p.p. into professional graduation, while technical graduation falls by 1.02 p.p. In that tercile, four of every five students private-voucher schools move into higher education land in an incomplete spell, and the share holding any degree does not rise (−0.27 p.p., statistically zero, in Panel B). In the top tercile vouchers moves fewer students and their destinations are imprecise; professional dropout rises by 0.47 p.p. and professional graduation by 0.43 p.p., neither distinguishable from zero. Lastly, the share holding any degree rises in none of the three terciles.

Table 10: Schooling Effects of the Voucher Sector, by Ability

	low	middle	high
	(1)	(2)	(3)
A. Terminal schooling state			
not enrolled	-2.79***	-1.81***	-1.30***
	(0.47)	(0.35)	(0.50)
professional graduate	0.75***	0.74***	0.43
	(0.20)	(0.23)	(0.43)
professional dropout	1.92***	1.72***	0.47
	(0.27)	(0.26)	(0.43)
technical graduate	-1.02***	-1.16***	-0.13
	(0.28)	(0.23)	(0.35)
technical dropout	1.14***	0.51**	0.52
	(0.33)	(0.25)	(0.32)
B. Summary			
any higher education	2.79***	1.81***	1.30***
	(0.47)	(0.35)	(0.50)
professional	2.67***	2.46***	0.91*
	(0.33)	(0.31)	(0.52)
technical	0.12	-0.65**	0.39
	(0.41)	(0.32)	(0.45)
any degree	-0.27	-0.42	0.31
	(0.33)	(0.31)	(0.47)

Notes: The voucher-induced change in the share of students reaching each terminal schooling state, in percentage points, within terciles of the ability distribution. Standard errors in parentheses are from a parametric bootstrap (500 draws). *, **, ***: significant at 10%, 5%, 1%.

7.5 Decomposing the Effects of Vouchers

We proceed to further examine the interplay between the returns to schooling and the effects of attending a private-voucher high school, by separating the effects of vouchers on schooling with the effects of vouchers on induced schooling decisions.

Let S^1 and S^0 denote the terminal schooling level a student reaches under the private-voucher and public high school options, respectively, and $Y(s, d)$ her outcome in schooling level s under option d. Then, the relative gain from attending a private-voucher high school can be expressed

as,

$$Y^1 - Y^0 \;=\; \underbrace{\sum_s \left[\mathbf{1}\{S^1 = s\} - \mathbf{1}\{S^0 = s\}\right] Y(s,0)}_{\text{reallocation}} \;+\; \underbrace{\sum_s \mathbf{1}\{S^1 = s\}\left[Y(s,1) - Y(s,0)\right]}_{\text{within-state}}.$$

The first term refers to how much the private-voucher option induces students to achieve schooling level s, relative to public schools, valued at the outcome under the public option. We call this term "reallocation". The second term is the outcomes difference associated with schooling level s, weighted by the voucher-induced schooling decision of attaining s; we call this term "within-state". Taking expectations and letting $\Delta\pi_s = E\left(\mathbf{1}\{S^1 = s\}\right) - E\left(\mathbf{1}\{S^0 = s\}\right)$, or the voucher-induced change in the share of students reaching schooling level s, and $\bar{R}_s = E\left(Y(s,0)\right)$, or the average outcome (associated with the public school option) in that schooling level, we get,

$$\begin{aligned} E\left(Y^1 - Y^0\right) \;=\; & \underbrace{\sum_s \Delta\pi_s \bar{R}_s}_{\text{net flows times returns}} \;+\; \underbrace{\sum_s \mathrm{Cov}\big(\mathbf{1}\{S^1 = s\} - \mathbf{1}\{S^0 = s\},\, Y(s,0)\big)}_{\text{selection on gains}} \\ & + \underbrace{E\left(\sum_s \mathbf{1}\{S^1 = s\}\left[Y(s,1) - Y(s,0)\right]\right)}_{\text{within-state}}, \end{aligned}$$

where the first term values the net student flows at average returns, and the second term is the covariance between the voucher-induced flows to schooling level s and the gains at that schooling level; we call this term "selection on gains". In practice, we compute every term by averaging over the 500,000 simulated students, for each of whom the model delivers the terminal schooling level and the outcome under both the private-voucher and the public options.

Table 11 presents the results of the decomposion exercise just described. It displays the total effect, and its decomposition into the reallocation term, in turn decomposed into its net flows annd selection gains parts, and the within-state terms. Columns (1) shows the results for employment, and column (2) does analogously for log wages. Attending a private-voucher in lieu of a public high school raises the probability of holding a formal job by 1.08 p.p. and log wages by 0.063, and on both outcomes essentially none of the effect is explained by the schooling induced by private-voucher schools. The within-state term accounts for 0.064 of the wage effect, and the reallocation term is -0.001, statistically indistinguishable from zero, as are both of its parts. Of the employment effect, 1.01 p.p. arise within state and 0.07 p.p. through reallocation, the latter again indistinguishable from zero, although here it nets two parts with opposite sign: a negative net flows times returns of -0.15 p.p. and a positive selection on gains of 0.22 p.p. These results tell that private-voucher schools are effective in raising wages and formal employment, but the

schooling reallocation they cause contributes virtually nothing to either effect.

Table 11: Decomposing the Effect of a Voucher High School

	employment (p.p.)	log wage
	(1)	(2)
total effect	1.08***	0.063***
	(0.29)	(0.012)
reallocation across states	0.07	-0.001
	(0.04)	(0.002)
net flows times returns	-0.15***	-0.002
	(0.03)	(0.001)
selection on gains	0.22***	0.001
	(0.03)	(0.001)
within-state effect	1.01***	0.064***
	(0.29)	(0.012)

Notes: The effect of attending a private-voucher in lieu of a public high school is written as the sum of a reallocation term, which values the voucher-induced change in the distribution of terminal schooling states at the returns those states pay under the public branch, and a within-state term, which holds the schooling state fixed and asks what the voucher does to the outcome. The reallocation term is split further into net flows times average returns and a selection-on-gains covariance. Standard errors in parentheses are from a parametric bootstrap (500 draws). *, **, ***: significant at 10%, 5%, 1%.

8 Conclusions

We ask how private-voucher high schools shape the transition from school to work in Chile. To that end we use administrative data that follow into higher education and the labor market a cohort of students required to choose a new high school because their public primary school offered no secondary grades. We combine those data with value-added regressions and a sequential model of schooling decisions and outcomes that deliver a variety of treatment effect parameters. We find that vouchers improve these students' labor-market outcomes, by essentially increasing the returns to each terminal schooling level students may end up with.

Three findings support this conclusion. First, higher education is not a panacea in this setting; what it pays depends on the type of program as much as on completing it. A professional degree raises wages by 0.36 log points relative to a high-school diploma and a technical degree by 0.24, but an incomplete professional spell lowers wages by 0.23 log points and formal employment by 9.8 percentage points, while an incomplete technical spell raises wages by 0.06 log points and formal employment by 2.2 percentage points. The expected return to enrolling in a professional program is therefore only 0.019 log points, statistically indistinguishable from zero, against 0.140 for a technical program. The professional option is also the riskier one, both because completion risk accounts for more of its variance and because wages within its terminal states are far more

dispersed. Second, private-voucher high schools move students toward exactly that option. They raise higher-education enrollment by 1.97 percentage points, entirely on the professional track and drawn from students who would otherwise not have enrolled. The induced enrollment ends up mostly as dropout—professional dropout rises by 1.37 percentage points, and professional graduation by a smaller 0.64, while technical graduation falls by 0.77—so the additional enrollment leaves degree attainment no higher. Third, voucher schools raise wages by 0.063 log points and formal employment by 1.1 percentage points. Decomposing the wage effect, essentially all of it comes from higher pay within schooling states, while the reallocation across schooling states that voucher schools cause adds nothing we can detect; the employment effect likewise arises almost entirely within states.

Three questions seem to us the natural next ones. Whether students understand the completion risk they take on when they choose a professional program is an open empirical question, and one with clear policy relevance. If the professional track is chosen partly on optimistic beliefs, information may do more than subsidies. Whether the within-state wage advantage of voucher schools reflects instruction, networks, or unmeasured skills matters for whether it can be reproduced elsewhere. And whether the pattern we document—access expanding along the track in which an incomplete spell is costliest—generalizes to other settings in which school choice is expanded without a corresponding change in the higher-education options available is, we suspect, the question on which the external validity of these results turns.

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A Data

The following are the various administrative and survey data sets for Chile that we use in our analysis.

- *National standardized exams (SIMCE) for 8th graders, student-level, 2000.*
  These data provide information on students' test scores for four different subjects, verbal, mathematics, social sciences, and natural sciences. They also provide information on students' gender and grade.
- *8th grade SIMCE's questionnaire to parents and tutors, 2000.*
  These data consist in the responses to a survey that parents and tutors answer during the days when the national standardized tests are taken. The survey is voluntary, and non-response is not negligible; mother's education, for instance, is missing for 28% of the cohort and 17.5% of our estimation sample. It provides information on students' household size, house amenities, and time use, total number of books available in the household, household total monthly income, parents and tutors' time use, education, indigenous identification, occupation, health insurance, participation in social programs, reasons for the choice of the school, beliefs on the student's future educational attainment, satisfaction with the school, knowledge of school's average performance in standardized tests, total monthly expenses related to the student's education other than tuition, and school's admission criteria, tuition, and fees.
- *National standardized exams (SIMCE), school-level, 8th grade in 2000, 10th grade in 1998.*
  These data provide information on schools' average test scores for verbal, mathematics, and social and natural sciences, municipality, management type (public, private-voucher, private-non-voucher), socio-economic category of the population served by the school, urban status, and number of students taking the tests.
- *Registry of students, 2004-2008.*
  These data provide information on students' gender, date of birth, age, municipality of residence, type and level of education, grade, class, grade repetition status, special education status, and various characteristics of the school of attendance, such as municipality, management type, single/double shift schedule, and urban status.
- *Registry of schools, 2000, 2004-2009.*
  These data provide information on schools' municipality, management type, urban status, address, and type and level of education offered.
- *National standardized college admission exams (PSU), student-level, 2005-2010.*
  These data provide information on students' college admission test scores for four different

subjects: verbal, mathematics, social sciences, and natural sciences. These exams are not mandatory, but are essential in determining students' chances of attending higher education. The data also provide information on students' birth date, home address, level of education of the parents, occupation of the parents, high school attended, high school GPA, preference ranking of higher education institutions-majors, among others.

- *National socio-economic characterization household survey (CASEN), 2003, 2006, 2009, 2013.*
  The CASEN series of household surveys corresponds to the most important piece of socio-economic information in Chile. It is used to inform public polices, and contains extensive information on individuals' education, health, employment, etc. It is nationally representative at the municipality level.

- *Registry of students in higher education institutions, 2007-2016.*
  These data provide detailed information on students' enrollment in higher education institutions, including the institution, major, geographical location where the classes are taken, as well as students' age, gender, and birth date. We use the 2007-2013 records, and for 2005 and 2006 the enrollment records of the centralized college admission system, so that higher-education enrollment is observed from 2005 through 2013. We restrict entry to higher education to the seven years after our cohort left secondary school in 2004, keeping the 2012 and 2013 records only for students whose first enrollment, as recorded in the register's own first-year-of-enrollment field, took place in 2011 or earlier; a student recorded in any 2007-2011 register meets that condition by construction.

- *Registry of individuals completing a higher education degree, 2007-2016.*
  These data provide information on the students completing a higher education degree, by year. It also contains detailed information on the degree (major and higher education institution), the time needed to complete the degree, the official duration of the degree, the exact date when the degree was completed, as well as students' age, gender, and birth date. We use the 2007-2013 records, so that degrees are observed through 2013, the last year of the labor-market window. A degree counts for a student who entered higher education by 2011 under the seven-year rule described above, or who began the graduating program by 2011, so that a degree earned in 2012 or 2013 by an on-time entrant is retained while a student first enrolling after 2011 is neither an entrant nor a graduate in our data.

- *Unemployment Insurance (Seguro de Cesantía) administrative records, 2012-2013.*
  These data come from Chile's unemployment insurance system, administered by the *Administradora de Fondos de Cesantía* (AFC), which registers the monthly earnings and the employing firm of every salaried worker holding a formal private-sector contract. The records

available to us cover January 2012 to October 2013, with a few hundred individuals also recorded in November 2013. We use them to measure each individual's formal employment status (any recorded formal job over that window) and labor earnings (the taxable earnings recorded over the window, summed across months and employers and deflated month by month to November-2000 US dollars), when the typical student in our sample is 26 to 27 years old—roughly eight to nine years after high-school graduation. We convert the sum to an annual rate by dividing it by the 22 months of the window (23 for the few individuals with a November 2013 record) and multiplying by twelve, so that months without a formal job count as zero earnings. Individuals who never hold a formal-sector job are retained and treated as not employed, which defines the extensive (employment) margin of our labor-market outcomes.

B Additional Tables

This appendix collects the supporting tables. Table B.1 sets out the exact inclusion of the covariates and instruments in each equation of the model. Tables B.2–B.7 present the maximum-likelihood estimates discussed in Section 7, and Tables B.8–B.10 compare the simulated model with the data. Table B.11 reports the returns to education by mother's education.

Table B.1: Variables Used in the Empirical Analysis

		schooling choices				labor market	
	measurement system	high-school type (D_1)	college enrollment (D_2–D_3)	professional vs. technical (D_4–D_5)	college graduation (D_6–D_9)	earnings	employment
covariates							
male	Y	Y	Y	Y	Y	Y	Y
mother's education: 12 years	Y	Y	Y	Y	Y		
mother's education: more than 12 years	Y	Y	Y	Y	Y		
mother's education: missing	Y	Y	Y	Y	Y		
father's education: 12 years	Y	Y	Y	Y	Y		
father's education: more than 12 years	Y	Y	Y	Y	Y		
father's education: missing	Y	Y	Y	Y	Y		
broken home	Y	Y	Y	Y	Y		
broken home: missing	Y	Y	Y	Y	Y		
household income: CLP 100–200K	Y	Y	Y	Y	Y		
household income: more than CLP 200K	Y	Y	Y	Y	Y		
household income: missing	Y	Y	Y	Y	Y		
region: north (age 14)	Y	Y					
region: center (age 14)	Y	Y					
region: north (high school)			Y	Y	Y	Y	Y
region: center (high school)			Y	Y	Y	Y	Y
exclusion restrictions							
% voucher schools in municipality		Y					
local unemployment gap (high–low skill, 2003)			Y				
local wage gap (high–low skill, 2003)			Y				
college in municipality			Y				
professional college in municipality				Y			
local unemployment gap (high–low skill, 2009)					Y		
local wage gap (high–low skill, 2009)					Y		
local unemployment (2013)							Y
local wage (2013)							Y
unobserved heterogeneity							
factor	Y	Y	Y	Y	Y	Y	Y

Notes: This table shows the allocation of the covariates and exclusion restrictions across the equations of our model. The column headers are the measurement system (the four 8th-grade tests), the four schooling-choice nodes of Figure 1 (high-school type D_1; college enrollment D_2–D_3; the professional-vs.-technical choice D_4–D_5; and college graduation D_6–D_9), and the two labor-market outcomes (earnings and employment). A "Y" means that the variable enters the corresponding equation. Education, broken home, household income, and region enter as categorical indicators; the omitted categories are education below 12 years, non-broken home, household income below CLP 100K per month, and the southern region, and a missing-data category is included for each. Variables measured at age 14 (the primary-school stage) enter the measurement system and the high-school choice D_1; their high-school-stage analogues enter the remaining equations. The exclusion restrictions are the local availability of private-voucher schools (D_1); the local high–low-skill unemployment and wage gaps and the local availability of a college (D_2–D_3); the local availability of a professional college (D_4–D_5); the local labor-market gaps measured in 2009 (D_6–D_9); and the 2013 local unemployment and wage levels of the skill group of each terminal state, which enter the employment equations only. The factor is the unobserved (ability) heterogeneity.

Table B.2: Estimates - Measurement System

subject:	verbal		math		social sciences		natural sciences	
	coef.	std. err.	coef.	std. err.	coef.	std. err.	coef.	std. err.
male	-0.216	0.006	0.065	0.006	0.165	0.006	0.007	0.006
mother's education: 12 years	0.193	0.008	0.176	0.008	0.197	0.008	0.161	0.008
mother's education: more than 12 years	0.295	0.013	0.264	0.013	0.306	0.013	0.283	0.013
mother's education: missing	-0.164	0.016	-0.181	0.016	-0.153	0.016	-0.162	0.016
father's education: 12 years	0.120	0.008	0.114	0.008	0.139	0.008	0.113	0.008
father's education: more than 12 years	0.259	0.012	0.258	0.012	0.276	0.012	0.242	0.012
father's education: missing	-0.140	0.014	-0.160	0.014	-0.147	0.014	-0.123	0.015
broken home	0.019	0.007	0.009	0.007	0.021	0.007	0.013	0.007
broken home: missing	0.267	0.017	0.297	0.017	0.254	0.018	0.284	0.018
household income: CLP 100–200K	0.088	0.007	0.073	0.008	0.089	0.008	0.056	0.008
household income: more than CLP 200K	0.133	0.009	0.130	0.009	0.137	0.009	0.104	0.009
household income: missing	0.017	0.017	0.010	0.016	0.023	0.017	-0.008	0.017
region: north (age 14)	-0.061	0.009	-0.069	0.008	-0.068	0.009	-0.082	0.009
region: center (age 14)	-0.078	0.006	-0.082	0.006	-0.068	0.006	-0.090	0.006
constant	-0.141	0.008	-0.266	0.008	-0.353	0.008	-0.219	0.008
factor	1.000	-	0.931	0.004	0.946	0.004	0.980	0.004
$\log(\sigma)$	-0.677	0.003	-0.552	0.003	-0.566	0.002	-0.586	0.003
observations	94,430		94,430		94,430		94,430	

This table reports the estimated coefficients of the maximum-likelihood measurement system: the four 8th-grade SIMCE tests (verbal, math, social sciences, natural sciences), all measured at age 14. The factor loading for verbal is normalized to one. The four test scores are normalized to mean zero and standard deviation one in the population of test takers; $\log(\sigma)$ is the natural logarithm of the standard deviation of the corresponding test's distribution. Estimates obtained by maximum likelihood (first estimation step). Omitted categories: mother's/father's education <12 years, non-broken home, household income below CLP 100K per month, and the southern region. Household income brackets are monthly figures. The factor is the unobserved (ability) heterogeneity; its loading is reported in the *factor* row.

Table B.3: Estimates - Schooling Decisions D_1, D_2, D_3

conditional on:			$D_1 = 1$		$D_1 = 0$	
decision node:	D_1: voucher school		D_2: enroll college		D_3: enroll college	
	coef.	std. err.	coef.	std. err.	coef.	std. err.
male	0.107	0.009	-0.091	0.015	-0.161	0.011
mother's education: 12 years	0.073	0.012	0.438	0.021	0.467	0.017
mother's education: more than 12 years	0.120	0.021	0.673	0.040	0.712	0.032
mother's education: missing	0.023	0.024	-0.393	0.042	-0.493	0.031
father's education: 12 years	-0.000	0.012	0.347	0.021	0.366	0.016
father's education: more than 12 years	0.022	0.019	0.595	0.036	0.591	0.028
father's education: missing	0.064	0.021	-0.286	0.037	-0.309	0.028
broken home	-0.015	0.011	-0.082	0.019	-0.076	0.015
broken home: missing	-0.130	0.028	0.400	0.050	0.447	0.037
household income: CLP 100–200K	0.117	0.012	0.231	0.020	0.283	0.015
household income: more than CLP 200K	0.207	0.015	0.631	0.026	0.574	0.020
household income: missing	0.071	0.027	0.267	0.048	0.311	0.035
region: north (age 14)	-0.455	0.014	-	-	-	-
region: center (age 14)	0.163	0.010	-	-	-	-
region: north (high school)	-	-	0.016	0.031	-0.129	0.016
region: center (high school)	-	-	0.020	0.018	-0.171	0.013
% voucher schools in municipality	0.894	0.018	-	-	-	-
local unemployment gap (high–low skill)	-	-	-0.419	0.162	-0.055	0.108
local wage gap (high–low skill)	-	-	-0.005	0.003	0.002	0.002
college in municipality	-	-	0.063	0.016	0.085	0.012
constant	-1.049	0.014	-0.441	0.027	-0.401	0.019
factor	0.038	0.007	0.565	0.012	0.605	0.009
observations	94,430		32,844		61,586	

This table reports the estimated coefficients for schooling decision nodes D_1, D_2, and D_3 in Figure 1. D_1 is the choice of a private-voucher (vs. public) high school; D_2 and D_3 are the decision to enroll in college, conditional on attending a private-voucher ($D_1 = 1$) or public ($D_1 = 0$) high school. Covariates for D_1 are measured at age 14; covariates for D_2 and D_3 at the high-school stage. Estimates obtained by maximum likelihood (first estimation step). Omitted categories: mother's/father's education <12 years, non-broken home, household income below CLP 100K per month, and the southern region. Household income brackets are monthly figures. The factor is the unobserved (ability) heterogeneity; its loading is reported in the *factor* row.

Table B.4: Estimates - Schooling Decisions D_4, D_5

conditional on:	$D_2 = 1$		$D_3 = 1$	
decision node:	D_4: prof. degree		D_5: prof. degree	
	coef.	std. err.	coef.	std. err.
male	-0.174	0.021	-0.062	0.016
mother's education: 12 years	0.224	0.026	0.257	0.021
mother's education: more than 12 years	0.343	0.042	0.463	0.034
mother's education: missing	-0.274	0.065	-0.238	0.052
father's education: 12 years	0.161	0.026	0.188	0.021
father's education: more than 12 years	0.416	0.040	0.377	0.031
father's education: missing	-0.114	0.060	-0.295	0.048
broken home	-0.067	0.027	-0.062	0.021
broken home: missing	0.318	0.071	0.351	0.055
household income: CLP 100–200K	0.089	0.030	0.148	0.022
household income: more than CLP 200K	0.315	0.034	0.352	0.026
household income: missing	0.054	0.071	0.086	0.054
region: north (high school)	0.070	0.040	0.071	0.022
region: center (high school)	-0.088	0.025	-0.087	0.018
professional college in municipality	0.013	0.022	0.048	0.016
constant	-0.086	0.034	-0.301	0.023
factor	0.454	0.016	0.566	0.013
observations	16,484		28,217	

This table reports the estimated coefficients for schooling decision nodes D_4 and D_5 in Figure 1: the choice of a professional (vs. technical) college degree, conditional on having enrolled in college from a private-voucher (D_4) or public (D_5) high school. The dependent variable equals one for a professional degree and zero for a technical degree. Estimates obtained by maximum likelihood (first estimation step). Omitted categories: mother's/father's education <12 years, non-broken home, household income below CLP 100K per month, and the southern region. Household income brackets are monthly figures. The factor is the unobserved (ability) heterogeneity; its loading is reported in the *factor* row.

Table B.5: Estimates - Schooling Decisions D_6, D_7, D_8, D_9

conditional on:	$D_4 = 1$		$D_4 = 0$		$D_5 = 1$		$D_5 = 0$	
decision node:	D_6: degree		D_7: degree		D_8: degree		D_9: degree	
	coef.	std. err.	coef.	std. err.	coef.	std. err.	coef.	std. err.
male	-0.310	0.027	-0.434	0.031	-0.330	0.021	-0.453	0.023
mother's education: 12 years	0.091	0.033	0.029	0.040	0.108	0.025	0.088	0.031
mother's education: more than 12 years	0.140	0.046	0.178	0.069	0.142	0.037	0.137	0.056
mother's education: missing	-0.194	0.102	-0.123	0.094	-0.108	0.087	-0.128	0.067
father's education: 12 years	0.018	0.034	0.055	0.040	0.013	0.026	0.042	0.030
father's education: more than 12 years	0.122	0.044	0.010	0.067	0.081	0.035	0.002	0.050
father's education: missing	-0.225	0.094	-0.215	0.087	-0.393	0.082	-0.211	0.061
broken home	-0.170	0.035	-0.097	0.040	-0.126	0.028	-0.091	0.030
broken home: missing	0.210	0.095	0.189	0.106	0.144	0.073	0.165	0.076
household income: CLP 100–200K	0.056	0.042	-0.016	0.044	0.058	0.031	-0.003	0.031
household income: more than CLP 200K	0.150	0.045	0.100	0.051	0.085	0.034	0.015	0.039
household income: missing	0.051	0.097	0.089	0.104	0.122	0.074	0.092	0.074
region: north (high school)	-0.248	0.047	-0.174	0.064	-0.186	0.028	-0.229	0.033
region: center (high school)	-0.243	0.031	-0.089	0.036	-0.160	0.024	-0.112	0.026
local unemployment gap (high–low skill)	0.119	0.307	0.110	0.345	-0.209	0.197	-0.018	0.212
local wage gap (high–low skill)	0.003	0.005	-0.001	0.006	-0.007	0.004	-0.013	0.005
constant	-0.015	0.050	0.070	0.053	0.012	0.036	0.237	0.036
factor	0.276	0.020	0.293	0.025	0.245	0.016	0.241	0.019
observations	9,516		6,968		15,660		12,557	

This table reports the estimated coefficients for schooling decision nodes D_6–D_9 in Figure 1: graduating from college (earning a degree), conditional on the high-school type and degree-type path. D_6/D_7 are for private-voucher students pursuing a professional/technical degree ($D_4 = 1/0$); D_8/D_9 are for public students pursuing a professional/technical degree ($D_5 = 1/0$). Estimates obtained by maximum likelihood (first estimation step). Omitted categories: mother's/father's education <12 years, non-broken home, household income below CLP 100K per month, and the southern region. Household income brackets are monthly figures. The factor is the unobserved (ability) heterogeneity; its loading is reported in the *factor* row.

Table B.6: Estimates - Formal Employment

final schooling level:	professional graduate		professional dropout		technical graduate		technical dropout		not enrolled	
	coef.	std. err.	coef.	std. err.	coef.	std. err.	coef.	std. err.	coef.	std. err.
A. Private-voucher high school										
male	0.158	0.042	0.190	0.038	0.510	0.055	0.444	0.049	0.506	0.023
region: north (high school)	-0.026	0.071	-0.145	0.074	0.093	0.110	-0.079	0.104	-0.035	0.050
region: center (high school)	0.196	0.046	0.282	0.047	0.357	0.061	0.094	0.060	0.126	0.028
local unemployment (own skill group)	-0.692	0.265	-0.412	0.340	0.545	0.338	-0.765	0.369	-0.449	0.178
local wage (own skill group)	0.046	0.074	0.272	0.119	0.172	0.101	0.395	0.163	0.262	0.073
constant	0.294	0.484	-1.176	0.680	-0.780	0.653	-1.488	0.924	-0.908	0.413
factor	-0.065	0.031	-0.087	0.028	0.078	0.043	0.067	0.039	0.038	0.019
observations	4,342		5,174		2,928		4,040		16,360	
B. Public high school										
male	0.213	0.032	0.270	0.029	0.538	0.039	0.558	0.037	0.560	0.016
region: north (high school)	0.185	0.043	0.098	0.047	0.197	0.055	0.065	0.062	0.017	0.028
region: center (high school)	0.259	0.035	0.187	0.035	0.229	0.040	0.209	0.042	0.092	0.018
local unemployment (own skill group)	-0.054	0.169	-0.137	0.230	-0.061	0.176	-0.697	0.291	-0.198	0.125
local wage (own skill group)	-0.009	0.052	0.232	0.082	0.087	0.063	0.305	0.108	0.200	0.048
constant	0.426	0.339	-1.079	0.466	-0.210	0.405	-1.143	0.616	-0.622	0.272
factor	-0.057	0.024	-0.134	0.022	0.026	0.030	0.052	0.029	0.009	0.014
observations	7,425		8,235		5,797		6,760		33,369	

This table reports the estimated coefficients of the ten formal-employment equations of the model, the equations for I_s^E in Section 5, one for each terminal schooling state s. Each column is a final schooling level and the two panels are the two types of high school, so that each column of a panel is one terminal state. The dependent variable equals one if the student holds a formal job at some point over the 2012–2013 window. The variance of the employment error is normalized to one, so the coefficients are those of a probit index. The local unemployment rate is the 2013 municipality unemployment rate and the local wage is the log of mean monthly earnings, both computed over residents aged 18 to 35 of the skill group of the state, those holding a higher-education degree in the four graduate states and those without one in the other six, so the constants are not comparable between the two groups of states. Estimates obtained by maximum likelihood (second estimation step); the standard errors are corrected for the first-stage estimation (Murphy and Topel, 1985). The omitted region category is the southern region, and the regions are those of residence at the high-school stage. The factor is the unobserved (ability) heterogeneity; its loading is reported in the *factor* row.

Table B.7: Estimates - Log Wages

final schooling level:	professional graduate		professional dropout		technical graduate		technical dropout		not enrolled	
	coef.	std. err.	coef.	std. err.	coef.	std. err.	coef.	std. err.	coef.	std. err.
A. Private-voucher high school										
male	0.190	0.047	0.325	0.046	0.484	0.047	0.513	0.038	0.640	0.020
region: north (high school)	0.189	0.078	0.160	0.078	0.407	0.100	0.402	0.074	0.173	0.039
region: center (high school)	0.108	0.052	0.412	0.055	0.384	0.053	0.223	0.044	0.104	0.023
constant	8.036	0.054	7.273	0.056	7.657	0.053	7.520	0.047	7.493	0.024
factor	0.181	0.036	-0.027	0.035	0.153	0.037	0.099	0.032	0.186	0.017
$\log(\sigma)$	0.267	0.011	0.327	0.010	0.124	0.011	0.097	0.009	0.130	0.004
observations	3,251		3,705		2,321		3,389		13,139	
B. Public high school										
male	0.157	0.037	0.397	0.039	0.502	0.037	0.569	0.031	0.740	0.015
region: north (high school)	0.267	0.049	0.407	0.052	0.316	0.051	0.337	0.042	0.268	0.021
region: center (high school)	0.119	0.042	0.212	0.046	0.110	0.040	0.106	0.035	-0.016	0.017
constant	8.037	0.038	7.210	0.043	7.677	0.036	7.495	0.033	7.416	0.016
factor	0.131	0.029	-0.027	0.030	0.186	0.031	0.108	0.026	0.116	0.012
$\log(\sigma)$	0.288	0.009	0.371	0.009	0.164	0.009	0.136	0.008	0.158	0.003
observations	5,261		5,504		4,299		5,500		26,312	

This table reports the estimated coefficients of the ten log-wage equations of the model, the equations for W_s in Section 5, one for each terminal schooling state s. Each column is a final schooling level and the two panels are the two types of high school, so that each column of a panel is one terminal state. The dependent variable is the natural logarithm of annual labor earnings over the 2012–2013 window, in November-2000 US dollars, and is observed only for the formally employed, so the observation counts are those of the employed students in each state. $\log(\sigma)$ is the natural logarithm of σ_s, the standard deviation of the idiosyncratic error of the state's log-earnings equation; the unobserved factor contributes to the dispersion of wages separately, through the loading reported in the *factor* row. Estimates obtained by maximum likelihood (second estimation step); the standard errors are corrected for the first-stage estimation (Murphy and Topel, 1985). The omitted region category is the southern region, and the regions are those of residence at the high-school stage. The factor is the unobserved (ability) heterogeneity; its loading is reported in the *factor* row.

Table B.8: Goodness of Fit - Measurement System

	mean		std. dev.	
	actual	model	actual	model
verbal	-0.141	-0.142	0.909	0.905
math	-0.141	-0.141	0.905	0.901
social sciences	-0.147	-0.147	0.914	0.910
natural sciences	-0.138	-0.138	0.915	0.911

Notes: This table compares our simulated model with the actual data. Simulations consist in 500,000 draws taken from the model and its estimates.

Table B.9: Goodness of Fit - Schooling Decisions

node	decision	actual	model
D_1	private-voucher high school	0.348	0.348
D_2	enroll in college | voucher	0.502	0.503
D_3	enroll in college | public	0.458	0.461
D_4	professional degree | voucher, enrolled	0.577	0.575
D_5	professional degree | public, enrolled	0.555	0.551
D_6	college degree | voucher, professional	0.456	0.453
D_7	college degree | voucher, technical	0.420	0.425
D_8	college degree | public, professional	0.474	0.467
D_9	college degree | public, technical	0.462	0.462

Notes: This table compares the actual data with our model's predictions for the schooling-decision nodes of Figure 1. The model column is based on 500,000 simulated observations from the model and its estimates. Each entry is the mean of the decision indicator over the indicated conditioning subsample.

Table B.10: Goodness of Fit - Labor-Market Outcomes

	formal employment		log annual wage		
	actual	model	actual	model	observations
A. By terminal schooling state					
private-voucher high school					
professional graduate	0.749	0.741	8.284	8.268	4,342
professional dropout	0.716	0.716	7.747	7.734	5,174
technical graduate	0.793	0.784	8.217	8.210	2,928
technical dropout	0.839	0.836	8.036	8.031	4,040
not enrolled	0.803	0.803	7.921	7.923	16,360
public high school					
professional graduate	0.709	0.707	8.268	8.261	7,425
professional dropout	0.668	0.669	7.619	7.611	8,235
technical graduate	0.742	0.743	8.012	8.021	5,797
technical dropout	0.814	0.812	7.952	7.954	6,760
not enrolled	0.789	0.787	7.862	7.859	33,369
B. By 8th-grade test score					
bottom quartile	0.784	0.781	7.808	7.809	23,608
second quartile	0.786	0.780	7.894	7.885	23,607
third quartile	0.774	0.769	7.975	7.966	23,608
top quartile	0.734	0.743	8.057	8.056	23,607
all students	0.770	0.768	7.932	7.928	94,430

Notes: This table compares our simulated model with the actual data for the two labor-market outcomes, formal employment and log annual wages. The model estimates a separate employment equation and a separate wage equation for each of the ten terminal schooling states of Figure 1, so each row of Panel A compares a distinct pair of equations with the data. The four 8th-grade test scores enter those equations only through the unobserved factor and never directly, so Panel B compares an ability gradient the outcome equations are not fitted on; its quartiles are cut on the average of the four scores in the estimation sample, and the same cuts are applied to the simulated scores. Wages are observed only for the formally employed, and the wage columns are means over them. Earnings are deflated to November-2000 US dollars and annualized over the 2012-2013 window, as described in Appendix A. The observations column counts students in the estimation sample (N=94,430, of whom 72,681 hold a formal job); the model columns are computed on 500,000 simulated observations.

Table B.11: Returns to Education by Mother's Education

	all	no degree	$\geq$ HS degree
	(1)	(2)	(3)
A. Employment			
professional graduate	-0.049***	-0.048***	-0.051***
	(0.006)	(0.006)	(0.005)
technical graduate	-0.015***	-0.015***	-0.014***
	(0.005)	(0.005)	(0.005)
professional dropout	-0.098***	-0.098***	-0.097***
	(0.005)	(0.005)	(0.005)
technical dropout	0.022***	0.022***	0.024***
	(0.004)	(0.004)	(0.004)
B. Log wages			
professional graduate	0.363***	0.372***	0.359***
	(0.024)	(0.024)	(0.023)
technical graduate	0.235***	0.236***	0.237***
	(0.020)	(0.020)	(0.020)
professional dropout	-0.231***	-0.225***	-0.233***
	(0.023)	(0.023)	(0.022)
technical dropout	0.061***	0.059***	0.067***
	(0.018)	(0.018)	(0.018)

Notes: Estimated average treatment effects (ATE) of each final schooling level relative to a high school diploma, on employment (Panel A) and log annual wages (Panel B). Columns (2) and (3) split students by their mother's education, those whose mother has no high-school degree (fewer than 12 years of schooling) and those whose mother has at least a high-school degree (12 years or more). Point estimates are from the model; standard errors in parentheses are from a parametric bootstrap (500 draws of the parameters from their estimated asymptotic distribution). *, **, ***: significant at 10%, 5%, 1%.